APPENDIX D
AIR QUALITY ANALYSIS AND SUPPORTING INFORMATION

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SUB-APPENDIX D.1
3400 LAGUNA STREET AIR QUALITY AND HEALTH RISK ASSESSMENT AND METHODOLOGY

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RIVERSIDE
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SAN LUIS OBISPO

MEMORANDUM

DATE: October 25, 2023

TO: Jeanie Poling, Senior Planner/San Francisco Planning

FROM: Cara Cunningham, Associate
Theresa Wallace, AICP, Principal

SUBJECT: Scope of Work Memorandum for the 3400 Laguna Street/Heritage on the Marina Project (Case File No. 2022-009819PPA)

INTRODUCTION

LSA Associates, Inc. (LSA) is under contract to prepare an Environmental Impact Report (EIR) for the proposed 3400 Laguna Street/Heritage on the Marina Project (project). Per the San Francisco Planning Department's (department's) Air Quality Screening analysis (dated July 5, 2023), the proposed project would not exceed the criteria air pollutant screening levels in the Bay Area Air Quality Management District (BAAQMD) CEQA Guidelines; therefore, a quantitative assessment of criteria air pollutant impacts from project construction and operation would not be required. However, due to the construction activity as well as the proximity of nearby sensitive receptors and the project site location within the City's Air Pollutant Exposure Zone (APEZ), LSA understands that a construction health risk assessment (HRA) would be required for the proposed project. LSA has prepared this memorandum to outline a proposed scope of work, methodology, and data assumptions for review and approval for the preparation of the construction HRA. Our proposed approach and scope of work is based on a review of the project site plans and aerial map of the project site, and constitutes our understanding of the reports required by the department.

PROJECT UNDERSTANDING

The approximately 68,090-square-foot (approximately 1.6-acre) project site at 3400 Laguna Street is located on a corner lot southeast of the Laguna Street and Bay Street intersection in the Marina neighborhood. The site is bounded by Bay Street to the north, single- and multi-family residences near to and along Octavia Street to the east, Francisco Street to the south, and Laguna Street to the west.

The project site is occupied by the Heritage on the Marina residential care retirement community, which consists of five existing structures of approximately 83,200 gross square feet: the Julia Morgan Building, the Perry Building, the Perry Building Connector, the Health Center, and the Caretaker's Cottage.

With development of the proposed project, the residential care facility would continue to operate on the project site. The proposed project would result in the demolition of two of the five existing

buildings (the Perry Connector and the Health Care Center) and construction of two new buildings (the Bay Building and the Francisco Building). The proposed project would result in the renovation of the other three existing buildings on the site. In total, the proposed project would add approximately 58,380 square feet of institutional use, and increase the number of residential care suites by 23 from 86 to 109.

Construction of the project would take approximately 29 months and is currently anticipated to occur between January 2027 and June 2029. The proposed new buildings would be constructed on mat foundations, and no impact or vibratory pile driving techniques would be used. The proposed project would require the excavation of 9,060 cubic yards of soil to a maximum depth of 15 feet. The total area of soil disturbance would be 29,750 square feet. Construction is scheduled for Monday through Friday, with no nighttime construction to occur.

Construction activity would use diesel-fueled equipment and would generate truck trips, which could increase concentrations of particulate matter and could expose nearby sensitive receptors to toxic air contaminants. Due to the construction activity as well as the proximity of the adjacent sensitive receptors and project site location within the City's APEZ, LSA understands that an HRA would be required.

SCOPE OF WORK

LSA will prepare three documents that together would comprise the construction HRA: 1) this scope of work memorandum; 2) a methodology memorandum; and 3) a results memorandum, which would include a list of modeling uncertainties/any conservative parameters used. The HRA will also include a project description and the following methodology and data assumptions.

Project Description and Receptors Analyzed

LSA will prepare a project description that will describe types and sources of pollutants that may be emitted from construction of the proposed project. The setting section will identify sensitive receptors and off-site workers and their proximity to the project site. The setting section will include graphics indicating the location of the project site and nearby receptor locations. Day care and school locations within 1,000 feet of the project site will also be identified.

Construction HRA

The following tasks will be conducted as part of the construction HRA:

- Confirm the receptor grid for construction prior to modeling, which will follow the Citywide Health Risk Assessment. Receptor heights, following those shown in the Citywide Health Risk Assessment technical support document, are shown in AERMOD output file data, which will be included as an appendix to the Methodology Memorandum.

- Coordinate with the department regarding the approach for the construction analysis.

- Construction emission sources will be modeled using utilizing BAAQMD, CAPCOA, and Environmental Planning Division guidance.

- Run the AERMOD air dispersion program. This will allow the use of the HARP health risk assessment post-processor to automatically process air dispersion modeling results that have been predicted using a unitized emission rate.

- Document the analysis and present the results, including mapping of the risk results, in the Construction HRA Results Memorandum.

If the department's thresholds are exceeded, LSA will work with the department to develop a set of practical, effective, and enforceable control measures to reduce identified impacts to a less-than-significant level, where possible. Control measures could include Tier 4 construction equipment, equipment electrification, and incorporating diesel particulate filters. All control measures will be quantified and reported in the Construction HRA Report. The results report will include methodology undertaken to determine the effectiveness of control measures.

SCHEDULE

Based on the City's review schedule timelines, LSA estimates that the HRA can be completed within timelines shown in Table 1, below. LSA will provide each document in clean and track changes versions.

Table 1: Proposed HRA Schedule

Milestone	Responsible Party	Weeks to Complete
Air Quality/Health Risk Assessment Scope of Work-1	LSA	1 week
Review of Air Quality/Health Risk Assessment Scope of Work-1	City	2 weeks
Revised Air Quality/Health Risk Assessment Scope of Work-2	LSA	3 days
Review of Air Quality/Health Risk Assessment Scope of Work-2	City	1 week
Final Air Quality/Health Risk Assessment Scope of Work	LSA	2 days
Air Quality/Health Risk Assessment Methodology Memo-1	LSA	1 week
Review of Air Quality/Health Risk Assessment Methodology Memo-1	City	4 weeks
Revised Air Quality/Health Risk Assessment Methodology Memo-2	LSA	1 week
Review of Air Quality/Health Risk Assessment Methodology-2	City	4 weeks
Final Air Quality/Health Risk Assessment Methodology Memo	LSA	1 week
Impact Preview Meeting with Air Quality Team	LSA/City	3 weeks
Draft Air Quality/Health Risk Assessment Results Memo-1	LSA	1 week
Review Draft Air Quality/Health Risk Assessment Results Memo-1	City	2 weeks
Revised Air Quality/Health Risk Assessment Results Memo-2	LSA	1 week
Review Air Quality/Health Risk Assessment Results Memo-2	City	1 week
Final Air Quality/Health Risk Assessment Modeling and Memo	LSA	1 week

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MEMORANDUM

DATE: March 20, 2024

TO: Megan Calpin, Senior Environmental Planner/San Francisco Planning

FROM: Cara Cunningham, Associate

SUBJECT: HRA Methodology Memorandum for the 3400 Laguna Street/Heritage on the Marina Project (Case File No. 2022-009819PPA)

INTRODUCTION

LSA Associates, Inc. (LSA) is under contract to prepare an Environmental Impact Report (EIR) for the proposed 3400 Laguna Street/Heritage on the Marina Project (project). Per the department's Air Quality Screening analysis (dated July 5, 2023), due to the construction activity as well as the proximity of nearby sensitive receptors and project site location within the City's Air Pollutant Exposure Zone (APEZ), LSA understands that a construction health risk assessment (HRA) would be required. LSA has prepared this memorandum to outline the methodology and data assumptions for review and approval for the preparation of the construction HRA.

The project's potential health risk impacts will be calculated using the California Emissions Estimator Model version 2022.1 (CalEEMod) and AERMOD. Model input assumptions were made using the information provided by the project sponsor and model defaults. LSA's proposed approach and scope of work is based on a review of the project site plans and aerial map of the project site, and constitutes our understanding of the requirements of the San Francisco Planning Department.

PROPOSED PROJECT

The approximately 68,005-square-foot (approximately 1.6-acre) project site at 3400 Laguna Street is located on a corner lot southeast of the Laguna Street and Bay Street intersection in the Marina neighborhood. The site is bounded by Bay Street to the north, single- and multi-family residences near to and along Octavia Street to the east, Francisco Street to the south, and Laguna Street to the west.

The project site is occupied by the Heritage on the Marina residential care retirement community, which consists of five existing structures of approximately 83,200 gross square feet: the Julia Morgan Building, the Perry Building, the Perry Building Connector, the Health Center, and the Caretaker's Cottage.

With development of the proposed project, the residential care facility would continue to operate on the project site. The proposed project would result in the demolition of two of the five existing

buildings (the Perry Connector and the Health Care Center) and construction of two new buildings (the Bay Building and the Francisco Building). The proposed project would result in the renovation of the other three existing buildings on the site. In total, the proposed project would add approximately 58,380 square feet of institutional use, and increase the number of residential care suites by 23 from 86 to 109. The basement level of the new buildings would include 31 vehicular parking spaces, including one car-share space, and 18 bicycle parking spaces. In addition, the proposed project would replace an existing 25-year-old diesel emergency generator with a new Tier 4 diesel emergency generator of similar size.[1] Any change in emissions associated with the new generator would likely be lower than those associated with the existing generator and are not accounted for in this HRA.

Construction of the project would take approximately 29 months and is currently anticipated to occur between January 2027 and June 2029. The proposed new buildings would be constructed on mat foundations, and no impact or vibratory pile driving techniques would be used. The proposed project would require the excavation of 9,060 cubic yards of soil to a maximum depth of 15 feet. The total area of soil disturbance would be 29,750 square feet. Construction is scheduled for Monday through Friday**,** with no nighttime construction to occur.

The project characteristics are shown in Table A.

Table A: Project Characteristics

Land Use	Unit Amount	Lot Acreage	Square Feet	Landscape Area Square Feet
Residential Congregate Care (Assisted Living)	109 units	1.6	141,580	3,870
Enclosed Parking with Elevator	31 spaces	0	12,400*	0

* CalEEMod default data

METHODOLOGY AND DATA ASSUMPTIONS

Construction Modeling

Construction activities associated with the proposed project would generate emissions associated with demolition, site preparation, grading, soil hauling, and other construction activities on the project site. Construction equipment exhaust would also be a source of air pollution. Construction emissions will be estimated using the latest version of CalEEMod. CalEEMod is the latest air quality emissions model developed by the California Air Pollution Control Officers Association (CAPCOA) and recommended by the Bay Area Air Quality Management District (BAAQMD). CalEEMod incorporates California Air Resources Board (CARB) approved Off-Road and On-Road Mobile-Source Emission Factor models (OFFROAD and EMFAC, respectively) and is designed to estimate emissions for land use development projects. CalEEMod allows for the input of project-specific information and includes default assumptions for when project-specific information is not available. Emissions

[1] Charles Pankow Builders, LTD, Generator Letter, October 18, 2023.

modeling input data and assumptions for the project schedule, off-road construction equipment, grading and demolition activities, construction worker and truck trips, and asphalt paving are presented below. Where project-specific data is not available, default assumptions from CalEEMod will be used to estimate project emissions.

Anticipated Schedule

The project's construction schedule and phasing were based on project-specific data provided by the project sponsor and is shown in Table B.

Table B: Construction Schedule

Phase	Start Date	End Date	Days/Week	Total Days
Demolition	1/4/2027	3/1/2027	5	40
Site Preparation	1/4/2027	3/1/2027	5	40
Grading	2/4/2027	4/29/2027	5	60
Building Construction	4/6/2027	11/2/2027	5	150
Architectural Coating & Finishing	9/6/2027	7/7/2028	5	220
Paving	5/6/2028	6/30/2028	5	40

Internal building improvements are anticipated to continue through June 2029.

Off-Road Equipment

The project's off-road diesel-fueled construction equipment that will be used in the model are based on project-specific data provided by the project sponsor. Construction equipment that will be used in this analysis are shown in Table C.

Table C: Construction Off-road Equipment

Equipment Type	Unit Amount	Hours/Day	Horsepower (HP)	Load Factor
Demolition				
Rubber Tired Dozers	1	8	367	0.4
Crawler Tractors	1	8	300	0.43
Excavators	1	8	273	0.38
Rubber Tired Loaders	1	8	600	0.36
Sweepers/Scrubbers	1	8	36	0.46
Site Preparation*				
Graders	1	8	148	0.41
Tractors/Loaders/ Backhoes	1	8	84	0.37
Rubber Tired Dozers	1	7	367	0.4
Grading*				
Graders	1	8	148	0.41
Rubber Tired Dozers	1	8	367	0.4
Tractors/Loaders/ Backhoes	2	7	84	0.37
Building Construction				
Aerial Lifts	4	8	46	0.31

Table C: Construction Off-road Equipment

Equipment Type	Unit Amount	Hours/Day	Horsepower (HP)	Load Factor
Air Compressors	1	8	49	0.48
Bore/Drill Rigs	1	8	83	0.5
Cement and Mortar Mixers	1	8	10	0.56
Forklifts	1	8	82	0.2
Generator Sets	1	8	14	0.74
Pumps	1	8	11	0.74
Rough Terrain Forklifts	1	8	130	0.4
Sweepers/Scrubbers	1	8	36	0.46
Paving*				
Cement and Mortar Mixers	1	6	10	0.56
Pavers	1	6	81	0.42
Rollers	1	7	36	0.38
Tractors/Loaders/Backhoes	1	8	84	0.37
Paving Equipment	1	8	89	0.36
Architectural Coating*				
Air Compressors	1	6	37	0.48

* CalEEMod default data

Grading and Demolition

The proposed project would require approximately 9,060 cubic yards off-haul and demolition of the 14,000 square feet of existing buildings, which will be included in the CalEEMod analysis. In addition, this analysis will assume compliance with the San Francisco Construction Dust Control Ordinance; therefore, construction dust emissions will be excluded from the analysis.

Construction Trips

Construction trips and trip lengths were based on CalEEMod defaults and are shown in Table D.

Table D: Construction Trips and Trip Lengths

Phase	# One-Way Trips Worker/Day	# One-Way Trips Vendor/Day	# One-Way Trips Hauling/Day	Trip Length Worker (mi)	Trip Length Vendor (mi)	Trip Length Hauling (mi)
Demolition	12.5	0	4	11.7	8.4	20.0
Site Preparation	7.5	0	0	11.7	8.4	20.0
Grading	10	0	19	11.7	8.4	20.0
Building Construction	83.7	13.7	0	11.7	8.4	20.0
Paving	12.5	0	0	11.7	8.4	20.0
Architectural Coating & Finishing	16.7	0	0	11.7	8.4	20.0

* CalEEMod default data

On-Road Fugitive Dust

On-road fugitive dust assumptions in CalEEMod were based on default assumptions and are shown in Table E.

Table E: On-Road Fugitive Dust Assumptions

Phase	% Pave Worker	% Pave Vendor	% Pave Hauling	Road Silt Loading (g/m2)	Material Silt Content %	Material Moisture Content %	Average Vehicle Weight (tons)	Mean Vehicle Speed (mph)
Demolition	100	100	100	0.1	8.5	0.5	2.4	40
Grading	100	100	100	0.1	8.5	0.5	2.4	40
Building Construction	100	100	100	0.1	8.5	0.5	2.4	40
Paving	100	100	100	0.1	8.5	0.5	2.4	40
Architectural Coating	100	100	100	0.1	8.5	0.5	2.4	40

* CalEEMod default data

Based on the CalEEMod emission results, LSA will prepare a construction HRA, which will evaluate construction-period health risks to on-site and off-site receptors resulting from emissions from the proposed project due to diesel PM, and $PM_{2.5}$. The construction HRA will analyze $PM_{2.5}$ concentrations, cancer risk, and non-cancer chronic hazard index (chronic). The BAAQMD and California Air Pollution Control Officers Association (CAPCOA) have prepared detailed guidance for the air modeling required in HRAs. San Francisco also has prepared a Citywide Health Risk Assessment and technical support document modeling health risk throughout the city. LSA will follow these documents and analyses in completing this task. This assessment will be conducted using the air dispersion modeling performed using the United States Environmental Protection Agency (USEPA) dispersion model AERMOD. The model provides a detailed estimate of exhaust

concentrations based on site and source geometry, source emissions strength, distance from the source to the receptor, and meteorological data.

Health Risk Modeling

The HRA will be prepared using technical information and HRA guidance and protocol from the BAAQMD,[1] the CARB,[2] the California Office of Environmental Health Hazard Assessment (OEHHA),[3] and the 2020 Citywide HRA.[4] This HRA will evaluate the estimated incremental increase in lifetime cancer risks from exposure to emissions of diesel particulate matter (DPM) associated with combustion (i.e., exhaust) and the annual average $PM_{2.5}$ concentrations associated with combustion and fugitive sources including tire wear, brake wear, and resuspended road dust, that would be emitted by project-related construction sources. While DPM is a complex mixture of gases and fine particles that includes over 40 substances that are listed by the United States Environmental Protection Agency (USEPA) as hazardous air pollutants and by CARB as a toxic air contaminant, in accordance with HRA modeling guidelines, the HRA will use PM_{10} emissions as a surrogate for DPM emissions.[5, 6] Pollutant concentrations will be estimated using the USEPA dispersion model AERMOD. The model provides a detailed estimate of exhaust concentrations based on site and source geometry, source emissions strength, distance from the source to the receptor, and meteorological data.

Consistent with the 2020 Citywide HRA, health risks from DPM and annual average $PM_{2.5}$ concentrations will be estimated at all sensitive receptors and off-site workers located within 1,000 feet of the proposed project's boundaries.

Additionally, the HRA will estimate existing plus project lifetime excess cancer risk and annual average $PM_{2.5}$ concentrations at the project's maximally exposed individual (MEI) by considering the

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1 Bay Area Air Quality Management District, Recommended Methods For Screening and Modeling Local Risks and Hazards, 2022, https://www.baaqmd.gov/~/media/files/planning-and-research/ceqa/ceqa-guidelines-2022/appendix-e-recommended-methods-for-screening-and-modeling-local-risks-and-hazards_final-pdf.pdf?rev=b8917a27345a4a629fc18fc8650951e4&sc_lang=en, accessed November 2023.

2 California Air Resources Board, Consolidated Table of OEHHA/ARB Approved Risk Assessment Health Values, last updated December 9, 2022, https://ww2.arb.ca.gov/sites/default/files/classic//toxics/healthval/contable.pdf, accessed October 2023.

3 Office of Environmental Health Hazard Assessment, Air Toxics Hot Spots Program Guidance Manual for the Preparation of Health Risk Assessments, February 2015, http://oehha.ca.gov/air/hot_spots/hotspots2015.html, accessed October 2023.

4 San Francisco Department of Public Health, and San Francisco Planning Department, The San Francisco Citywide Health Risk Assessment: Technical Support Documentation, September 2020.

5 Office of Environmental Health Hazard Assessment, For the "Proposed Identification of Diesel Exhaust as a Toxic Air Contaminant" Part B: Health Risk Assessment for Diesel Exhaust, May 1998 https://www.arb.ca.gov/toxics/dieseltac/part_b.pdf, accessed October 2023.

6 BAAQMD, Regulation 2 Permits Rule 5 New Source Review of Toxic Air Contaminants, 2016a, December 7, https://www.baaqmd.gov/~/media/dotgov/files/rules/reg-2-rule-5-new-source-review-of-toxic-air-contaminants/documents/rg0205_120716-pdf.pdf?la=en&rev=c1b53fb7e7bd41d3b9522796d3c4ccff, accessed October 2023.

proposed project's impact in aggregate with existing sources. The existing sources are attributable to the other mobile and stationary sources as modeled in the 2020 Citywide HRA.

Toxic Air Contaminant Concentrations

Inputs to the model will include general modeling parameters that account for atmospheric conditions, emission rates for each contaminant from the project sources, source parameters that characterize the activities generating emissions, variable phase durations to characterize construction schedule, and receptor characteristics (e.g., resident child, school-age child, childcare facility).

General AERMOD Parameters. General AERMOD modeling parameters are presented in Table F. Meteorological data from the Mission Bay (Site ID# 23234) monitoring site will be used. Digital terrain data from the Shuttle Radar Topography Mission (SRTM) will be used to assigned terrain heights every 20 meters, consistent with the receptor grid spacing that will be used in the air dispersion modeling. The SRTM data provides full coverage of the US in 1 by 1 degree blocks with an approximate resolution of 30 by 30 meters. Terrain elevation data was imported from Lakes Environmental and the AERMAP software will be used to process the digital terrain data into a format compatible with AERMOD.

Table F: Overall AERMOD Modeling Parameters

Pathway	Characteristic	Parameter
Control Pathway	Averaging Time	Period average
	Urban vs Rural	Urban
	Model Version	AERMOD v. 22119
Source Pathway	Spacing	See Table G
	Release Height	See Table G
	Initial Vertical Dimension	See Table G
	Initial Lateral Dimension	See Table G
Receptor Pathway	Receptor Height	1.5 meter
	Grid	20 m x 20 meters
Meteorological Pathway	Surface Data	Mission Bay (Site ID# 23234) monitoring site
	Upper Air	Oakland International Airport (Site ID# 23230)
	Station Elevation	2.4 meters

Sources:
Bay Area Air Quality Management District, Air Quality Guidelines Appendix E: Recommended Methods for Screening and Modeling Local Risks and Hazards, August 2023, https://www.baaqmd.gov/~/media/files/planning-and-research/ceqa/ceqa-guidelines-2022/appendix-e-recommended-methods-for-screening-and-modeling-local-risks-and-hazards_final-pdf.pdf?rev=b8917a27345a4a629fc18fc8650951e4&sc_lang=en, accessed October 2023.
2. San Francisco Department of Public Health, and San Francisco Planning Department, The San Francisco Citywide Health Risk Assessment: Technical Support Documentation, September 2020.

Emission Rates. Emission rates of DPM along with exhaust and fugitive[1] $PM_{2.5}$ from the various emission sources (e.g., construction equipment) will be based on the anticipated hours of activity for

[1] Fugitive emissions include brake wear and tire wear obtained from EMFAC2021 and entrained paved road dust derived from Bay Area Air Quality Management District, Miscellaneous Process Methodology 7.9

each source and other information as described above. It will be assumed that all haul trucks and all vendor trucks are diesel heavy heavy-duty trucks (HHDT).

Each source will be modeled with a unitized emission rate of 1 gram/second (g/s). The modeled concentration at each receptor (micrograms per cubic meter [μ/m^3]/[g/s]) represents a "dispersion factor." The dispersion factor from each source will then be multiplied by its annual average emission rate to determine the annual average ambient pollutant concentration at every receptor from that source. Each source's resulting pollutant concentrations will be added together at each receptor to obtain the final result.

Annual average ambient air concentrations will be determined by multiplying the annual average dispersion factors by the annual average emission rates for each source. For simplicity, the model assumed a constant annual emission rate for each year.

Source Parameters. Table G presents AERMOD source configurations and parameters that will be used in the model. Construction sources will be modeled as an area source within AERMOD using the same release parameters used in the 2020 Citywide HRA, including a release height of 5 meters and an initial vertical dimension of 1.4 meters.

Haul truck and vendor trips associated with construction will be modeled as line-volume sources. The line-volume source width corresponds to the roadway width, while the modeled release height was 2.55 meters and the initial vertical dimension was 2.37 meters, consistent with the 2020 Citywide HRA modeling. Road dust generated from the truck trips will be modeled with a release height at ground level and an initial vertical dimension of 1 meter. This analysis only accounts for the locally emitted onsite emissions from the haul truck and vendor trips.

Onsite idling during construction from haul truck and vendor deliveries will be modeled as an area source with the same release height and initial vertical dimensions as the haul truck and vendor trips, which are 2.55 meters and 2.37 meters, respectively.[1]

Table G: AERMOD Source Modeling Parameters

Parameter	Off-Road Construction Equipment	On-Road Trucks	On-Road Vehicles	Mobile Sources	Loading Truck Idling
Source Type	Area	Line Volume	Line Volume	Line Volume	Point
Source Dimension	Project Area	Variable	Variable	Variable	n/a
Number of Sources	1	Variable	Variable	Variable	n/a
Release Height (m)	5	2.55	1.70	2.0	3.84
Initial Vertical Dimension (m)	1.4	2.37	1.58	2.3	n/a
Initial Lateral Dimension (m)	n/a	Variable	Variable	Variable	n/a

Entrained Road Travel, Paved Road Dust, March 2018, https://ww3.arb.ca.gov/ei/areasrc/fullpdf/full7-9_2018.pdf, accessed October 2023.

[1] This assumption reduces total fugitive $PM_{2.5}$ emissions to identify the $PM_{2.5}$ emissions that are locally emitted if considering full worker, vendor, and haul trips lengths, some of which are greater than 20 miles.

Table G: AERMOD Source Modeling Parameters

Parameter	Off-Road Construction Equipment	On-Road Trucks	On-Road Vehicles	Mobile Sources	Loading Truck Idling
Stack Diameter (m)	n/a	n/a	n/a	n/a	0.1
Temperature (K)	n/a	n/a	n/a	n/a	366
Velocity (m/s)	n/a	n/a	n/a	n/a	25
Hours per Day	9 (8am – 5pm)	9 (8am – 5pm)	9 (8am – 5pm)	9 (8am – 5pm)	15 min/hr/truck
Days per Week	5 (M-F)	5 (M-F)	5 (M-F)	5 (M-F)	5 (M-F)

Source: San Francisco Department of Public Health, and San Francisco Planning Department, The San Francisco Citywide Health Risk Assessment: Technical Support Documentation, September 2020.

Receptors. To represent on-site and off-site sensitive receptors and off-site workers near the project site, a modeling grid with receptors placed every 20-meters will be input to extend 1,000 feet from the project boundary in AERMOD; this is consistent with the receptor grid used in the 2020 Citywide HRA. Receptors will be placed at a height of 1.5 meters above terrain height, which represents the default breathing height for ground-floor receptors (i.e., human residents). Sensitive receptor locations include residential areas (based on residential land use and/or zoning data), childcare centers, and schools (for children under 16 years of age). Sensitive receptors and off-site workers are included in the receptor grid shown Figure 1.

Figure 1 – Project Site and Receptor Grid



Health Risks

Health risks from project-related sources will be estimated for construction activity including off-road construction equipment, paving, haul truck travel and idling, vendor trucks, and worker trips. The CARB Hotspots Analysis & Reporting Program (HARP) 2 Air Dispersion Modeling and Risk Tool will be used to calculate the health risks for the proposed project.

Pollutants Modeled. The TACs that will be modeled in the health risk assessment include DPM, modeled using exhaust PM_{10} as a surrogate, and $PM_{2.5}$ exhaust from off-road equipment, diesel truck travel and idling, and fugitive $PM_{2.5}$ from brake wear, tire wear, and entrained road dust from on-road truck travel on paved roads.

Exposure Assessment. Receptor types evaluated in the HRA will include existing on-site and off-site resident, childcare, school student, and worker receptors. Figure 1 above shows the location of the receptors included in the analysis.

Exposure Assessment. All exposure assumptions are presented in Table H and are based on OEHHA and BAAQMD risk assessment guidelines.

Table H: Exposure Parameters

Receptor Type	Age Group	Daily Breathing Rate (L/kg day or L/kg 8 hours)	Exposure Duration (years)	Fraction of Time at Home	Exposure Frequency (days per year)	Averaging Time (days)	Model Adjustment Factor	Age Sensitivity Factor
On-site Resident	Seniors	240	2.4	1	350	25,550	1	1
Off-site Resident	Third Trimester	240	0.25	1	350	25,550	1	10
	Age 0-2 Years	1,200	2	1	350	25,550	1	10
	Age 2-16 Years	520	2.4	1	350	25,550	1	3
	16-30	240	2.4	0.73	350	25,550	1	1
Off-site Childcare	Age 0-2 Years	1,200	2	n/a	350	25,550	4.2	10
	Age 2-9 Years	640	2.4	n/a	350	25,550	4.2	3
Off-site School	Age 2-16 Years	520	2.4	n/a	350	25,550	4.2	3
On-site and Off-site Workers	Age 16-41 Years	170	2.4	n/a	250	9,125	1	1

Sources:
1. Office of Environmental Health Hazard Assessment, Air Toxics Hot Spots Program Guidance Manual for the Preparation of Health Risk Assessments, February 2015.
http://oehha.ca.gov/air/hot_spots/hotspots2015.html, accessed October 2023.
2. San Francisco Department of Public Health, and San Francisco Planning Department, The San Francisco Citywide Health Risk Assessment: Technical Support Documentation, September 2020.

Table H: Exposure Parameters

Receptor Type	Age Group	Daily Breathing Rate (L/kg day or L/kg 8 hours)	Exposure Duration (years)	Fraction of Time at Home	Exposure Frequency (days per year)	Averaging Time (days)	Model Adjustment Factor	Age Sensitivity Factor

3. Bay Area Air Quality Management District, Air Quality Guidelines Appendix E: Recommended Methods for Screening and Modeling Local Risks and Hazards, August 2023. https://www.baaqmd.gov/~/media/files/planning-and-research/ceqa/ceqa-guidelines-2022/appendix-e-recommended-methods-for-screening-and-modeling-local-risks-and-hazards_final-pdf.pdf?rev=b8917a27345a4a629fc18fc8650951e4&sc_lang=en, accessed October 2023.

Toxicity Assessment. The toxicity values used in the analysis for DPM will be from CARB/OEHHA.[1] This toxicity value is for carcinogenic (cancer) effects; the primary pathway for exposures is assumed to be inhalation. The incremental risks were determined for each TAC emission source and summed to obtain an estimated total incremental cancer health risk. Table I presents this value.

Table I: Carcinogenic Toxicity Values

Chemical/TAC	EMFAC Gasoline TOG Speciation (% TOG)	Inhalation Cancer Potency Factors (mg/kg-day)$^{-1}$	Weighted Inhalation Cancer Potency Factor
Diesel Exhaust (Diesel Vehicles)			
Diesel Particulate Matter	n/a	1.1	n/a
TOG Exhaust (Gasoline Vehicles)			
Acetaldehyde	0.28%	1.0E-02	2.80E-05
Benzene	2.47%	1.0E-01	2.47E-03
1,3-Butadiene	0.55%	6.0E-01	3.30E-03
Ethylbenzene	1.05%	8.7E-03	9.14E-05
Formaldehyde	1.58%	2.1E-02	3.32E-04
Naphthalene	0.05%	1.2E-01	6.00E-05
Weighted Total Toxicity Factor	—	—	6.28e-03
TOG Evaporative (Gasoline Vehicles)			
Benzene	0.36%	0.1	3.60E-04
Ethylbenzene	0.12%	8.70E-03	1.04E-05
Weighted Total Toxicity Factor	—	—	3.70E-04

Sources:
1. Bay Area Air Quality Management District, Air Quality Guidelines Appendix E: Recommended Methods for Screening and Modeling Local Risks and Hazards, August 2023. Available at: https://www.baaqmd.gov/~/media/files/planning-and-research/ceqa/ceqa-guidelines-2022/appendix-e-recommended-methods-for-screening-and-modeling-local-risks-and-hazards_final-pdf.pdf?rev=b8917a27345a4a629fc18fc8650951e4&sc_lang=en. Accessed October 2023.
2. California Air Resources Board. 2018. Consolidated Table of OEHHA/ARB Approved Risk Assessment Health Values. Last Updated: August 20, 2018. Available at: http://www.arb.ca.gov/toxics/healthval/contable.pdf. Accessed October 2023.

[1] California Air Resources Board, Consolidated Table of OEHHA/ARB Approved Risk Assessment Health Values. Last Updated: October 2, 2020, https://ww2.arb.ca.gov/sites/default/files/classic//toxics/healthval/contable.pdf, accessed October 2023.

Age Sensitivity Factors. Cancer risk estimates will be weighted by a factor of 10 for exposures that occur from the third trimester of pregnancy to 2 years of age and by a factor of three for exposures that occur from 2 years through 16 years of age.

Modeling Adjustment Factors. For exposure to construction emissions, since construction represents a non-continuous source, a modeling adjustment factor will be used for any school or childcare receptors to determine the long-term average daily concentration the school or childcare sensitive receptor may be breathing during their time at childcare or school. This is consistent with OEHHA (2015) protocol.

Construction is scheduled to occur Monday through Friday, from 7:00 a.m. to 8:00 p.m. Nighttime construction is not expected. This analysis assumes that construction activities will occur during an 8-hour period to coincide with when children are present in schools and childcare.

For school and childcare sensitive receptors, a model adjustment factor of 4.2 will be used (equation = [7 days / 5 days] * [24-hour residential exposure / 8 hours of exposure while at school/childcare] = 4.2).

Calculation of Intake. The dose estimated for each exposure pathway is a function of the concentration of a chemical and the intake of that chemical. The intake factor for inhalation will be calculated as follows using Equation 1.

$$\textbf{Equation 1: } IF_{inh} = \frac{DBR * FAH * EF * ED * MAF * ASF * CF}{AT}$$

Where:

IF_{inh} = Intake Factor for Inhalation (m^3/kg-day)
DBR = Daily Breathing Rate (L/kg-day)
FAH = Frequency of time at home (unitless)
EF = Exposure Frequency (days/year)
ED = Exposure Duration (years)
AT = Averaging Time (days)
MAF = Model Adjustment Factor (unitless)
ASF = Age Sensitivity Factor (unitless)
CF = Conversion Factor, 0.001 (m^3/L)

Calculation of Cancer Risk. Excess lifetime cancer risk is estimated as the upper-bound incremental probability that an individual will develop cancer over a lifetime as a direct result of exposure to carcinogens. Excess lifetime cancer risk is expressed as a unitless probability, and is calculated as the number of cancer incidences per million individuals. The cancer risk for each chemical is calculated by multiplying the chemical intake or dose at the human exchange boundaries (e.g., lungs) by the

chemical-specific cancer potency factor. Excess lifetime cancer risk occurs exclusively through the inhalation pathway and is calculated according to Equation 2, below.

Equation 2: $Risk_{inh} = C_i * IF_{inh} * CPF_i * CF_1 * CF_2$

Where:

$Risk_{inh}$ = Cancer risk; the incremental probability of an individual developing cancer as a result of inhalation exposure to a particular carcinogen (per million)

C_i = Average annual air concentration of chemical, from AERMOD (μ/m^3)

IF_{inh} = Intake Factor for Inhalation (m^3/kg-day)

CPF_i = Cancer potency factor for chemical (mg chemical/kg body weight-day)$^{-1}$

CF_1 = Conversion factor, micrograms to milligrams (mg/µg)

CF_2 = Risk per million individuals

i = Chemical

Risk from Existing Sources. The HRA will estimate the lifetime excess cancer risk and annual average $PM_{2.5}$ concentrations from existing TAC sources within 1,000 feet of the proposed project based on the Citywide HRA.

Risk Analysis. This analysis will evaluate the existing plus proposed project cancer risk and annual average $PM_{2.5}$ concentrations from the proposed project in order to identify the maximum lifetime excess cancer risks and annual average $PM_{2.5}$ concentration contributions from the proposed project and determine the project's contribution to existing health risks in the area.

Cumulative Analysis

The HRA will also qualitatively evaluate cumulative health risks associated with the proposed project and the background sources in the surrounding area at the on-site and off-site sensitive receptor locations. Cumulative development in the project vicinity includes only the Marina Improvement and Remediation Project. While it is not expected that the two projects would be constructed at the same time, cancer risk is the probability of getting cancer over a 30 year (or 25-year for workers) exposure period, and cumulative project emissions don't need to occur at the same time to combine to result in a cumulative cancer risk impact. Thus, the HRA will qualitatively evaluate the possibility of the project's cancer risk impact to increase impacts at the project MEI under cumulative conditions.

SUB-APPENDIX D.2
3400 LAGUNA STREET AIR QUALITY AND HEALTH RISK ASSESSMENT RESULTS

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CARLSBAD
CLOVIS
IRVINE
LOS ANGELES
PALM SPRINGS
POINT RICHMOND
RIVERSIDE
ROSEVILLE
SAN LUIS OBISPO

MEMORANDUM

DATE: May 13, 2024

TO: Megan Calpin, Senior Environmental Planner/San Francisco Planning

FROM: Cara Cunningham, Associate

SUBJECT: HRA Results Memorandum for the 3400 Laguna Street/Heritage on the Marina Project (Case File No. 2022-009819PPA)

INTRODUCTION

LSA Associates, Inc. (LSA) has prepared a construction Health Risk Assessment (HRA) for the proposed 3400 Laguna Street/Heritage on the Marina Project (project). This memorandum provides an analysis of the project's potential health risk impacts using the California Emissions Estimator Model version 2022.1 (CalEEMod) and AERMOD, which are attached to this memorandum. Emission calculation methodologies are discussed in the HRA Methodology Memorandum for the 3400 Laguna Street/Heritage on the Marina Project (Case File No. 2022-009819PPA) prepared by LSA, dated March 20, 2024.

PROPOSED PROJECT

The approximately 68,005-square-foot (approximately 1.6-acre) project site at 3400 Laguna Street is located on a corner lot southeast of the Laguna Street and Bay Street intersection in the Marina neighborhood. The site is bounded by Bay Street to the north, single- and multi-family residences near to and along Octavia Street to the east, Francisco Street to the south, and Laguna Street to the west.

The project site is occupied by the Heritage on the Marina residential care retirement community, which consists of five existing structures of approximately 83,200 gross square feet: the Julia Morgan Building, the Perry Building, the Perry Building Connector, the Health Center, and the Caretaker's Cottage.

With development of the proposed project, the residential care facility would continue to operate on the project site. The proposed project would result in the demolition of two of the five existing buildings (the Perry Connector and the Health Care Center) and construction of two new buildings (the Bay Building and the Francisco Building). The proposed project would result in the renovation of the other three existing buildings on the site. In total, the proposed project would add approximately 58,380 square feet of institutional use, and increase the number of residential care suites by 23 from 86 to 109. The basement level of the new buildings would include 31 vehicular parking spaces, including one car-share space, and 18 bicycle parking spaces. In addition, the proposed project would replace an existing 25-year-old diesel emergency generator with a new Tier

4 diesel emergency generator of similar size.[1] Any change in emissions associated with the new generator would likely be lower than those associated with the existing generator and are not accounted for in this HRA.

Construction of the project would take approximately 29 months and is currently anticipated to occur between January 2027 and June 2029. The proposed new buildings would be constructed on mat foundations, and no impact or vibratory pile driving techniques would be used. The proposed project would require the excavation of 9,060 cubic yards of soil to a maximum depth of 15 feet. The total area of soil disturbance would be 29,750 square feet. Construction is scheduled to occur Monday through Friday, with no nighttime construction to occur.

CONSTRUCTION HEALTH RISK ASSESSMENT RESULTS

The modeled excess cancer risk and annual $PM_{2.5}$ concentrations at the maximally exposed on-site residents, off-site residents, off-site childcare receptors, off-site school receptors, and on-site and off-site workers are shown in Tables A and B. Chronic Hazard Index (HI) resulting from the proposed project is also disclosed below in Table C. The maximally exposed receptor is the receptor location with the maximum cancer risk, $PM_{2.5}$ concentration, and chronic HI from the proposed project. Model snap shots of the sources are attached. Background health risk data in these tables are from the San Francisco Planning Department's 2020 Citywide HRA database.

Table A: Existing Plus Project Lifetime Cancer Risk at Maximally Exposed Individual Receptors

Cancer Risk	On-site Residential Receptor	Off-site Residential Receptor	Off-site Childcare Receptor	Off-site School Receptor	On-site Worker Receptor	Off-site Worker Receptor
Receptor coordinates (UTM X, UTM Y)	550120, 4184080	550140, 4184080	550240, 4184160	549760, 4183980	550120, 4184080	550240, 4184160
Proposed project cancer risk (per 1 million)	**1.37**	**26.31**	**8.44**	**0.14**	**0.69**	**0.05**
Existing lifetime excess cancer risk (2020)[1] (per 1 million)	108.16	107.64	133.99	95.39	108.16	133.99
Existing + proposed project cancer risk (per 1 million)	**109.53**	**133.95**	**142.43**	**95.53**	**108.85**	**134.04**

Source: LSA (February 2024).
Notes:
[1] Background cancer risk at receptor from the 2020 Citywide HRA database.
APEZ = Air Pollutant Exposure Zone
UTM = Universal Transverse Mercator
UTM X = eastward-measured distance
UTM Y = northward-measured distance

[1] Charles Pankow Builders, LTD, Generator Letter, October 18, 2023.

As shown in Table A, the maximum cancer risk from the proposed project would be approximately 26.31 in one million at the maximally exposed individual (MEI) off-site residential receptor. The following construction control measure would reduce the cancer risk at all receptor types analyzed:

- The project sponsor shall ensure all off-road diesel-powered construction equipment of 25 horsepower or more meets the California Air Resources Board Tier 4 Final emissions standards or equivalent.

Table B: Existing Plus Project Annual Average $PM_{2.5}$ Concentration at Maximally Exposed Individual Receptors

$PM_{2.5}$ Concentration	**On-site Residential Receptor**	**Off-site Residential Receptor**	**Off-site Childcare Receptor**	**Off-site School Receptor**	**On-site Worker Receptor**	**Off-site Worker Receptor**
Receptor coordinates (UTM X, UTM Y)	550120, 4184080	550140, 4184080	550240, 4184160	549760, 4183980	550120, 4184080	550240, 4184160
Proposed project annual average $PM_{2.5}$ concentration ($\mu g/m^3$)	**0.13**	**0.01**	**0.02**	**0.01**	**0.12**	**<0.01**
Existing annual average $PM_{2.5}$ concentration (2020)[1] ($\mu g/m^3$)	8.73	8.73	9.28	8.56	8.73	9.28
Existing + proposed project annual average $PM_{2.5}$ concentration ($\mu g/m^3$)	**8.86**	**8.74**	**9.30**	**8.57**	**8.85**	**9.28**

Source: LSA (February 2024).
Notes:
[1] Background $PM_{2.5}$ concentration at receptor from the 2020 Citywide HRA database.

APEZ = Air Pollutant Exposure Zone
UTM = Universal Transverse Mercator
UTM X = eastward-measured distance
UTM Y = northward-measured distance

As shown in Table B, the results of the analysis indicate that the maximum $PM_{2.5}$ concentration at the MEI would be 0.13 $\mu g/m^3$ at an on-site residential receptor.

Table C: Chronic Hazard Index at Maximally Exposed Individual Receptors

Chronic Hazard Index	**On-site Residential Receptor**	**Off-site Residential Receptor**	**Off-site Childcare Receptor**	**Off-site School Receptor**	**On-site Worker Receptor**	**Off-site Worker Receptor**
Receptor coordinates (UTM X, UTM Y)	550120, 4184080	550140, 4184080	550240, 4184160	549760, 4183980	550120, 4184080	550240, 4184160
Proposed Project Chronic Hazard Index	**0.06**	**0.03**	**<0.01**	**<0.01**	**0.03**	**<0.01**

Source: LSA (May 2024).
UTM = Universal Transverse Mercator
UTM X = eastward-measured distance
UTM Y = northward-measured distance

Furthermore, the proposed project would result in a maximum chronic HI of 0.06 at the MEI, which is the on-site residential receptor.

Tables D, E, and F identify the results of the analysis with implementation of the construction control measures discussed above.

Table D: Existing Plus Project Lifetime Cancer Risk at Maximally Exposed Individual Receptors with Control Measures

Cancer Risk	On-site Residential Receptor	Off-site Residential Receptor	Off-site Childcare Receptor	Off-site School Receptor	On-site Worker Receptor	Off-site Worker Receptor
Receptor coordinates (UTM X, UTM Y)	550120, 4184080	550140, 4184080	550240, 4184160	549760, 4183980	550120, 4184080	550240, 4184160
Proposed project cancer risk (per 1 million)	**0.29**	**5.49**	**1.76**	**0.03**	**0.14**	**0.01**
Existing lifetime excess cancer risk (2020)[1] (per 1 million)	108.16	107.64	133.99	95.39	108.16	133.99
Existing + proposed project cancer risk (per 1 million)	**108.45**	**113.13**	**135.75**	**95.42**	**108.30**	**134.0**

Source: LSA (February 2024).
Notes:
[1] Background cancer risk at receptor from the 2020 Citywide HRA database.
APEZ = Air Pollutant Exposure Zone
UTM = Universal Transverse Mercator
UTM X = eastward-measured distance
UTM Y = northward-measured distance

As shown in Table D, with the identified control measure, the maximum cancer risk from the project would be 5.49 in 1 million at the off-site residential receptor.

Table E: Existing Plus Project Annual $PM_{2.5}$ Concentration at the Maximally Exposed Individual Receptors with Control Measures

$PM_{2.5}$ Concentration	On-site Residential Receptor	Off-site Residential Receptor	Off-site Childcare Receptor	Off-site School Receptor	On-site Worker Receptor	Off-site Worker Receptor
Receptor coordinates (UTM X, UTM Y)	550120, 4184080	550140, 4184080	550240, 4184160	549760, 4183980	550120, 4184080	550240, 4184160
Proposed project annual average $PM_{2.5}$ concentration ($\mu g/m^3$)	**0.02**	**0.02**	**0.01**	**<0.01**	**0.03**	**<0.01**
Existing annual average $PM_{2.5}$ concentration (2020)[1] ($\mu g/m^3$)	8.73	8.73	9.28	8.56	8.73	9.28
Existing + proposed project annual average $PM_{2.5}$ concentration ($\mu g/m^3$)	**8.75**	**8.75**	**9.29**	**8.56**	**8.76**	**9.28**

Source: LSA (February 2024).
Notes:
[1] Background $PM_{2.5}$ concentration at receptor from the 2020 Citywide HRA database.
APEZ = Air Pollutant Exposure Zone
UTM = Universal Transverse Mercator
UTM X = eastward-measured distance
UTM Y = northward-measured distance

With implementation of the identified control measure, Table E shows the maximum $PM_{2.5}$ concentration from the proposed project would be 0.03 $\mu g/m^3$ at the on-site worker receptor.

Table F: Chronic Hazard Index at Maximally Exposed Individual Receptors with Control Measures

Chronic Hazard Index	On-site Residential Receptor	Off-site Residential Receptor	Off-site Childcare Receptor	Off-site School Receptor	On-site Worker Receptor	Off-site Worker Receptor
Receptor coordinates (UTM X, UTM Y)	550120, 4184080	550140, 4184080	550240, 4184160	549760, 4183980	550120, 4184080	550240, 4184160
Proposed project Chronic Hazard Index	**0.01**	**0.01**	**<0.01**	**<0.01**	**0.01**	**<0.01**

Source: LSA (May 2024).
UTM = Universal Transverse Mercator
UTM X = eastward-measured distance
UTM Y = northward-measured distance

In addition, as shown in Table F, with implementation of the identified control measure, the proposed project would result in a maximum chronic HI of 0.01 at the MEI, which includes the on-site residential receptor, off-site residential receptor, and on-site worker receptor.

CUMULATIVE HEALTH RISK ASSESSMENT

This section presents information regarding potential cumulative health risks in combination with the existing plus project health risks at the project MEI. Below is a list of cumulative projects; however, because of the lack of available emissions data for the cumulative projects, cumulative health risks were not evaluated quantitatively. Table G lists the cumulative projects and provides a brief description, the expected risk sources associated with each project, and the distances from the project site.

Table G: Health Risks from Project Construction to Off-Site Receptors

Location	Project Name and Description	Potential Source of Health Risk	Distance from the Project Site (feet)
San Francisco Marina	**Marina Improvement and Remediation Project:** This project would redevelop the San Francisco marina into a waterfront park and implement environmental remediation.	diesel particulate matter (DPM) and $PM_{2.5}$ from construction equipment	750 feet

Source: LSA (February 2024).

The cumulative project would add an additional health risk impact from diesel powered construction equipment, contributing to greater cumulative health risks; however, the proposed project's contribution to the cumulative health risk would remain the same as under existing plus project conditions.

Attachments: A: CalEEMod Output Sheets
B: HRA Model Snapshots

ATTACHMENT A

CALEEMOD OUTPUT SHEETS

3400 Laguna Street/Heritage on the Marina Project Custom Report

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8. User Changes to Default Data

1. Basic Project Information

1.1. Basic Project Information

Data Field	Value
Project Name	3400 Laguna Street/Heritage on the Marina Project
Construction Start Date	7/6/2026
Lead Agency	—
Land Use Scale	Project/site
Analysis Level for Defaults	County
Windspeed (m/s)	3.90
Precipitation (days)	41.8
Location	3400 Laguna St, San Francisco, CA 94123, USA
County	San Francisco
City	San Francisco
Air District	Bay Area AQMD
Air Basin	San Francisco Bay Area
TAZ	1029
EDFZ	1
Electric Utility	Pacific Gas & Electric Company
Gas Utility	Pacific Gas & Electric
App Version	2022.1.1.21

1.2. Land Use Types

Land Use Subtype	Size	Unit	Lot Acreage	Building Area (sq ft)	Landscape Area (sq ft)	Special Landscape Area (sq ft)	Population	Description
Congregate Care (Assisted Living)	109	Dwelling Unit	1.60	141,580	3,870	—	254	—

Enclosed Parking with Elevator	31.0	Space	0.00	12,400	0.00	—	—	—

1.3. User-Selected Emission Reduction Measures by Emissions Sector

Sector	#	Measure Title
Construction	C-5	Use Advanced Engine Tiers

2. Emissions Summary

2.1. Construction Emissions Compared Against Thresholds

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Un/Mit.	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Unmit.	10.5	25.5	33.4	0.06	0.86	4.00	4.86	0.79	1.65	2.44	—	7,412	7,412	0.57	0.35	7,535
Mit.	9.83	11.5	35.5	0.06	0.17	4.00	4.18	0.17	1.65	1.82	—	7,412	7,412	0.57	0.35	7,535
% Reduced	6%	55%	-6%	—	80%	—	14%	79%	—	25%	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Unmit.	10.5	43.4	47.2	0.10	1.73	6.12	7.85	1.59	2.72	4.32	—	12,296	12,296	0.80	0.39	12,432
Mit.	9.83	8.96	58.2	0.10	0.21	6.12	6.34	0.21	2.72	2.94	—	12,296	12,296	0.80	0.39	12,432
% Reduced	6%	79%	-23%	—	88%	—	19%	87%	—	32%	—	—	—	—	—	—
Average Daily (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Unmit.	3.46	10.4	13.5	0.02	0.35	1.21	1.56	0.33	0.47	0.79	—	3,047	3,047	0.18	0.10	3,083

Mit.	3.39	4.59	15.4	0.02	0.07	1.21	1.29	0.07	0.47	0.54	—	3,047	3,047	0.18	0.10	3,083
% Reduced	2%	56%	-14%	—	79%	—	18%	78%	—	32%	—	—	—	—	—	—
Annual (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Unmit.	0.63	1.89	2.46	< 0.005	0.06	0.22	0.29	0.06	0.09	0.15	—	504	504	0.03	0.02	510
Mit.	0.62	0.84	2.81	< 0.005	0.01	0.22	0.23	0.01	0.09	0.10	—	504	504	0.03	0.02	510
% Reduced	2%	56%	-14%	—	79%	—	18%	78%	—	32%	—	—	—	—	—	—

2.2. Construction Emissions by Year, Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Year	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily - Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	10.5	25.5	33.4	0.06	0.86	4.00	4.86	0.79	1.65	2.44	—	7,412	7,412	0.57	0.35	7,535
2028	9.67	4.98	8.50	0.01	0.16	0.24	0.40	0.15	0.06	0.21	—	1,375	1,375	0.05	0.01	1,380
Daily - Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	10.5	43.4	47.2	0.10	1.73	6.12	7.85	1.59	2.72	4.32	—	12,296	12,296	0.80	0.39	12,432
2028	9.21	0.84	1.59	< 0.005	0.02	0.14	0.15	0.01	0.03	0.05	—	269	269	0.01	0.01	271
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	3.28	10.4	13.5	0.02	0.35	1.21	1.56	0.33	0.47	0.79	—	3,047	3,047	0.18	0.10	3,083
2028	3.46	0.77	1.33	< 0.005	0.02	0.06	0.08	0.02	0.01	0.03	—	219	219	0.01	< 0.005	220
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	0.60	1.89	2.46	< 0.005	0.06	0.22	0.29	0.06	0.09	0.15	—	504	504	0.03	0.02	510
2028	0.63	0.14	0.24	< 0.005	< 0.005	0.01	0.02	< 0.005	< 0.005	0.01	—	36.3	36.3	< 0.005	< 0.005	36.4

2.3. Construction Emissions by Year, Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Year	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily - Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	9.83	11.5	35.5	0.06	0.17	4.00	4.18	0.17	1.65	1.82	—	7,412	7,412	0.57	0.35	7,535
2028	9.29	1.99	8.77	0.01	0.03	0.24	0.27	0.03	0.06	0.09	—	1,375	1,375	0.05	0.01	1,380
Daily - Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	9.83	8.96	58.2	0.10	0.21	6.12	6.34	0.21	2.72	2.94	—	12,296	12,296	0.80	0.39	12,432
2028	9.12	0.68	1.44	< 0.005	< 0.005	0.14	0.14	< 0.005	0.03	0.03	—	269	269	0.01	0.01	271
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	2.52	4.59	15.4	0.02	0.07	1.21	1.29	0.07	0.47	0.54	—	3,047	3,047	0.18	0.10	3,083
2028	3.39	0.40	1.32	< 0.005	< 0.005	0.06	0.06	< 0.005	0.01	0.02	—	219	219	0.01	< 0.005	220
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2027	0.46	0.84	2.81	< 0.005	0.01	0.22	0.23	0.01	0.09	0.10	—	504	504	0.03	0.02	510
2028	0.62	0.07	0.24	< 0.005	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	—	36.3	36.3	< 0.005	< 0.005	36.4

3. Construction Emissions Details

3.1. Demolition (2027) - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	2.35	17.5	18.2	0.05	0.70	—	0.70	0.64	—	0.64	—	5,725	5,725	0.23	0.05	5,745
Demolition	—	—	—	—	—	0.23	0.23	—	0.03	0.03	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.26	1.97	2.05	0.01	0.08	—	0.08	0.07	—	0.07	—	643	643	0.03	0.01	645
Demolition	—	—	—	—	—	0.03	0.03	—	< 0.005	< 0.005	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.05	0.36	0.37	< 0.005	0.01	—	0.01	0.01	—	0.01	—	106	106	< 0.005	< 0.005	107
Demolition	—	—	—	—	—	< 0.005	< 0.005	—	< 0.005	< 0.005	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.03	0.03	0.37	0.00	0.00	0.10	0.10	0.00	0.02	0.02	—	103	103	< 0.005	< 0.005	104
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

Hauling	0.01	0.55	0.47	< 0.005	< 0.005	0.08	0.08	< 0.005	0.02	0.03	—	315	315	0.07	0.05	332
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.04	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	11.6	11.6	< 0.005	< 0.005	11.7
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	< 0.005	0.06	0.05	< 0.005	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	—	35.3	35.3	0.01	0.01	37.3
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.01	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	1.92	1.92	< 0.005	< 0.005	1.94
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	—	5.85	5.85	< 0.005	< 0.005	6.17

3.2. Demolition (2027) - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.55	3.53	28.5	0.05	0.11	—	0.11	0.11	—	0.11	—	5,725	5,725	0.23	0.05	5,745
Demolition	—	—	—	—	—	0.23	0.23	—	0.03	0.03	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.06	0.40	3.20	0.01	0.01	—	0.01	0.01	—	0.01	—	643	643	0.03	0.01	645

Demolition	—	—	—	—	—	0.03	0.03	—	< 0.005	< 0.005	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.01	0.07	0.58	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	106	106	< 0.005	< 0.005	107
Demolition	—	—	—	—	—	< 0.005	< 0.005	—	< 0.005	< 0.005	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.03	0.03	0.37	0.00	0.00	0.10	0.10	0.00	0.02	0.02	—	103	103	< 0.005	< 0.005	104
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.01	0.55	0.47	< 0.005	< 0.005	0.08	0.08	< 0.005	0.02	0.03	—	315	315	0.07	0.05	332
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.04	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	11.6	11.6	< 0.005	< 0.005	11.7
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	< 0.005	0.06	0.05	< 0.005	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	—	35.3	35.3	0.01	0.01	37.3
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.01	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	1.92	1.92	< 0.005	< 0.005	1.94
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	—	5.85	5.85	< 0.005	< 0.005	6.17

3.3. Site Preparation (2027) - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	1.19	10.4	11.6	0.02	0.47	—	0.47	0.43	—	0.43	—	2,065	2,065	0.08	0.02	2,072
Dust From Material Movement	—	—	—	—	—	2.44	2.44	—	1.17	1.17	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.13	1.17	1.30	< 0.005	0.05	—	0.05	0.05	—	0.05	—	232	232	0.01	< 0.005	233
Dust From Material Movement	—	—	—	—	—	0.27	0.27	—	0.13	0.13	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.02	0.21	0.24	< 0.005	0.01	—	0.01	0.01	—	0.01	—	38.4	38.4	< 0.005	< 0.005	38.5
Dust From Material Movement	—	—	—	—	—	0.05	0.05	—	0.02	0.02	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.02	0.22	0.00	0.00	0.06	0.06	0.00	0.01	0.01	—	61.7	61.7	< 0.005	< 0.005	62.5
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.02	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	6.95	6.95	< 0.005	< 0.005	7.04
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	< 0.005	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	1.15	1.15	< 0.005	< 0.005	1.17
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.4. Site Preparation (2027) - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.19	1.01	11.9	0.02	0.04	—	0.04	0.04	—	0.04	—	2,065	2,065	0.08	0.02	2,072

Dust From Material Movement	—	—	—	—	—	2.44	2.44	—	1.17	1.17	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.02	0.11	1.33	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	232	232	0.01	< 0.005	233
Dust From Material Movement	—	—	—	—	—	0.27	0.27	—	0.13	0.13	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	< 0.005	0.02	0.24	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	38.4	38.4	< 0.005	< 0.005	38.5
Dust From Material Movement	—	—	—	—	—	0.05	0.05	—	0.02	0.02	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.02	0.22	0.00	0.00	0.06	0.06	0.00	0.01	0.01	—	61.7	61.7	< 0.005	< 0.005	62.5
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Worker	< 0.005	< 0.005	0.02	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	6.95	6.95	< 0.005	< 0.005	7.04
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	< 0.005	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	1.15	1.15	< 0.005	< 0.005	1.17
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.5. Grading (2027) - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	1.37	12.2	13.9	0.02	0.54	—	0.54	0.50	—	0.50	—	2,455	2,455	0.10	0.02	2,464
Dust From Material Movement	—	—	—	—	—	2.77	2.77	—	1.34	1.34	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	1.37	12.2	13.9	0.02	0.54	—	0.54	0.50	—	0.50	—	2,455	2,455	0.10	0.02	2,464
Dust From Material Movement	—	—	—	—	—	2.77	2.77	—	1.34	1.34	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.23	2.04	2.32	< 0.005	0.09	—	0.09	0.08	—	0.08	—	410	410	0.02	< 0.005	412
Dust From Material Movement	—	—	—	—	—	0.46	0.46	—	0.22	0.22	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.04	0.37	0.42	< 0.005	0.02	—	0.02	0.02	—	0.02	—	67.9	67.9	< 0.005	< 0.005	68.2
Dust From Material Movement	—	—	—	—	—	0.08	0.08	—	0.04	0.04	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.02	0.33	0.00	0.00	0.08	0.08	0.00	0.02	0.02	—	87.0	87.0	< 0.005	< 0.005	87.5
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.04	2.46	2.23	0.01	0.02	0.36	0.38	0.02	0.10	0.12	—	1,489	1,489	0.32	0.24	1,571
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.02	0.30	0.00	0.00	0.08	0.08	0.00	0.02	0.02	—	82.3	82.3	< 0.005	< 0.005	83.3
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.04	2.60	2.23	0.01	0.02	0.36	0.38	0.02	0.10	0.12	—	1,488	1,488	0.31	0.24	1,569
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.05	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	13.8	13.8	< 0.005	< 0.005	14.0

Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.01	0.43	0.37	< 0.005	< 0.005	0.06	0.06	< 0.005	0.02	0.02	—	249	249	0.05	0.04	262
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.01	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	2.28	2.28	< 0.005	< 0.005	2.31
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	< 0.005	0.08	0.07	< 0.005	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	—	41.2	41.2	0.01	0.01	43.4

3.6. Grading (2027) - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.23	1.20	14.2	0.02	0.05	—	0.05	0.05	—	0.05	—	2,455	2,455	0.10	0.02	2,464
Dust From Material Movement	—	—	—	—	—	2.77	2.77	—	1.34	1.34	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.23	1.20	14.2	0.02	0.05	—	0.05	0.05	—	0.05	—	2,455	2,455	0.10	0.02	2,464
Dust From Material Movement	—	—	—	—	—	2.77	2.77	—	1.34	1.34	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.04	0.20	2.38	< 0.005	0.01	—	0.01	0.01	—	0.01	—	410	410	0.02	< 0.005	412
Dust From Material Movement	—	—	—	—	—	0.46	0.46	—	0.22	0.22	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.01	0.04	0.43	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	67.9	67.9	< 0.005	< 0.005	68.2
Dust From Material Movement	—	—	—	—	—	0.08	0.08	—	0.04	0.04	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.02	0.33	0.00	0.00	0.08	0.08	0.00	0.02	0.02	—	87.0	87.0	< 0.005	< 0.005	87.5
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.04	2.46	2.23	0.01	0.02	0.36	0.38	0.02	0.10	0.12	—	1,489	1,489	0.32	0.24	1,571
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.02	0.30	0.00	0.00	0.08	0.08	0.00	0.02	0.02	—	82.3	82.3	< 0.005	< 0.005	83.3
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.04	2.60	2.23	0.01	0.02	0.36	0.38	0.02	0.10	0.12	—	1,488	1,488	0.31	0.24	1,569
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.05	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	13.8	13.8	< 0.005	< 0.005	14.0

Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.01	0.43	0.37	< 0.005	< 0.005	0.06	0.06	< 0.005	0.02	0.02	—	249	249	0.05	0.04	262
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.01	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	2.28	2.28	< 0.005	< 0.005	2.31
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	< 0.005	0.08	0.07	< 0.005	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	—	41.2	41.2	0.01	0.01	43.4

3.7. Building Construction (2027) - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	1.05	10.1	13.8	0.02	0.30	—	0.30	0.27	—	0.27	—	2,259	2,259	0.09	0.02	2,267
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	1.05	10.1	13.8	0.02	0.30	—	0.30	0.27	—	0.27	—	2,259	2,259	0.09	0.02	2,267
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.43	4.16	5.72	0.01	0.12	—	0.12	0.11	—	0.11	—	935	935	0.04	0.01	938
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.08	0.76	1.04	< 0.005	0.02	—	0.02	0.02	—	0.02	—	155	155	0.01	< 0.005	155
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.21	0.15	2.74	0.00	0.00	0.69	0.69	0.00	0.16	0.16	—	728	728	0.01	0.01	732
Vendor	0.01	0.60	0.44	< 0.005	< 0.005	0.10	0.10	< 0.005	0.03	0.03	—	393	393	0.05	0.06	413
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.21	0.18	2.49	0.00	0.00	0.69	0.69	0.00	0.16	0.16	—	689	689	0.01	0.03	697
Vendor	0.01	0.63	0.45	< 0.005	< 0.005	0.10	0.10	< 0.005	0.03	0.03	—	393	393	0.05	0.06	412
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.08	0.07	1.01	0.00	0.00	0.28	0.28	0.00	0.07	0.07	—	285	285	0.01	0.01	289
Vendor	0.01	0.26	0.18	< 0.005	< 0.005	0.04	0.04	< 0.005	0.01	0.01	—	163	163	0.02	0.02	171
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.01	0.18	0.00	0.00	0.05	0.05	0.00	0.01	0.01	—	47.3	47.3	< 0.005	< 0.005	47.9
Vendor	< 0.005	0.05	0.03	< 0.005	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	—	26.9	26.9	< 0.005	< 0.005	28.2
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.8. Building Construction (2027) - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.48	7.03	15.5	0.02	0.11	—	0.11	0.10	—	0.10	—	2,259	2,259	0.09	0.02	2,267
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.48	7.03	15.5	0.02	0.11	—	0.11	0.10	—	0.10	—	2,259	2,259	0.09	0.02	2,267
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.20	2.91	6.40	0.01	0.05	—	0.05	0.04	—	0.04	—	935	935	0.04	0.01	938
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.04	0.53	1.17	< 0.005	0.01	—	0.01	0.01	—	0.01	—	155	155	0.01	< 0.005	155
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.21	0.15	2.74	0.00	0.00	0.69	0.69	0.00	0.16	0.16	—	728	728	0.01	0.01	732
Vendor	0.01	0.60	0.44	< 0.005	< 0.005	0.10	0.10	< 0.005	0.03	0.03	—	393	393	0.05	0.06	413

Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.21	0.18	2.49	0.00	0.00	0.69	0.69	0.00	0.16	0.16	—	689	689	0.01	0.03	697
Vendor	0.01	0.63	0.45	< 0.005	< 0.005	0.10	0.10	< 0.005	0.03	0.03	—	393	393	0.05	0.06	412
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.08	0.07	1.01	0.00	0.00	0.28	0.28	0.00	0.07	0.07	—	285	285	0.01	0.01	289
Vendor	0.01	0.26	0.18	< 0.005	< 0.005	0.04	0.04	< 0.005	0.01	0.01	—	163	163	0.02	0.02	171
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.02	0.01	0.18	0.00	0.00	0.05	0.05	0.00	0.01	0.01	—	47.3	47.3	< 0.005	< 0.005	47.9
Vendor	< 0.005	0.05	0.03	< 0.005	< 0.005	0.01	0.01	< 0.005	< 0.005	< 0.005	—	26.9	26.9	< 0.005	< 0.005	28.2
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.9. Paving (2028) - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.43	4.13	6.47	0.01	0.15	—	0.15	0.13	—	0.13	—	991	991	0.04	0.01	995
Paving	0.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.05	0.45	0.71	< 0.005	0.02	—	0.02	0.01	—	0.01	—	109	109	< 0.005	< 0.005	109
Paving	0.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.01	0.08	0.13	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	18.0	18.0	< 0.005	< 0.005	18.1
Paving	0.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.03	0.02	0.39	0.00	0.00	0.10	0.10	0.00	0.02	0.02	—	107	107	< 0.005	< 0.005	107
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.04	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	11.1	11.1	< 0.005	< 0.005	11.1
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Worker	< 0.005	< 0.005	0.01	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	1.84	1.84	< 0.005	< 0.005	1.85
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.10. Paving (2028) - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.14	1.30	6.89	0.01	0.03	—	0.03	0.03	—	0.03	—	991	991	0.04	0.01	995
Paving	0.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.02	0.14	0.76	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	109	109	< 0.005	< 0.005	109
Paving	0.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	< 0.005	0.03	0.14	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	18.0	18.0	< 0.005	< 0.005	18.1
Paving	0.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.03	0.02	0.39	0.00	0.00	0.10	0.10	0.00	0.02	0.02	—	107	107	< 0.005	< 0.005	107
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.04	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	11.1	11.1	< 0.005	< 0.005	11.1
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.01	0.00	0.00	< 0.005	< 0.005	0.00	< 0.005	< 0.005	—	1.84	1.84	< 0.005	< 0.005	1.85
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.11. Architectural Coating & Finishing (2027) - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Off-Road Equipment	0.11	0.83	1.13	< 0.005	0.02	—	0.02	0.02	—	0.02	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.11	0.83	1.13	< 0.005	0.02	—	0.02	0.02	—	0.02	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.03	0.19	0.26	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	30.6	30.6	< 0.005	< 0.005	30.7
Architectural Coatings	2.08	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	< 0.005	0.03	0.05	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	5.06	5.06	< 0.005	< 0.005	5.08
Architectural Coatings	0.38	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.03	0.55	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	146	146	< 0.005	< 0.005	146
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.04	0.50	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	138	138	< 0.005	0.01	139
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.01	0.01	0.11	0.00	0.00	0.03	0.03	0.00	0.01	0.01	—	31.6	31.6	< 0.005	< 0.005	32.0
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.02	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	5.23	5.23	< 0.005	< 0.005	5.30
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.12. Architectural Coating & Finishing (2027) - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Off-Road Equipment	0.02	0.65	0.96	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.02	0.65	0.96	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	< 0.005	0.15	0.22	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	30.6	30.6	< 0.005	< 0.005	30.7
Architectural Coatings	2.08	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	< 0.005	0.03	0.04	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	5.06	5.06	< 0.005	< 0.005	5.08
Architectural Coatings	0.38	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.03	0.55	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	146	146	< 0.005	< 0.005	146
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.04	0.50	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	138	138	< 0.005	0.01	139
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.01	0.01	0.11	0.00	0.00	0.03	0.03	0.00	0.01	0.01	—	31.6	31.6	< 0.005	< 0.005	32.0
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.02	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	5.23	5.23	< 0.005	< 0.005	5.30
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.13. Architectural Coating & Finishing (2028) - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Off-Road Equipment	0.11	0.81	1.12	< 0.005	0.02	—	0.02	0.01	—	0.01	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.11	0.81	1.12	< 0.005	0.02	—	0.02	0.01	—	0.01	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.04	0.30	0.41	< 0.005	0.01	—	0.01	0.01	—	0.01	—	49.4	49.4	< 0.005	< 0.005	49.6
Architectural Coatings	3.35	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.01	0.05	0.08	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	8.18	8.18	< 0.005	< 0.005	8.20
Architectural Coatings	0.61	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.03	0.52	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	143	143	< 0.005	< 0.005	144
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.03	0.47	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	135	135	< 0.005	0.01	137
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.01	0.01	0.17	0.00	0.00	0.05	0.05	0.00	0.01	0.01	—	50.2	50.2	< 0.005	< 0.005	50.4
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.03	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	8.31	8.31	< 0.005	< 0.005	8.34
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

3.14. Architectural Coating & Finishing (2028) - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Location	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Onsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Off-Road Equipment	0.02	0.65	0.96	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.02	0.65	0.96	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	134	134	0.01	< 0.005	134
Architectural Coatings	9.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	0.01	0.24	0.36	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	49.4	49.4	< 0.005	< 0.005	49.6
Architectural Coatings	3.35	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Off-Road Equipment	< 0.005	0.04	0.07	< 0.005	< 0.005	—	< 0.005	< 0.005	—	< 0.005	—	8.18	8.18	< 0.005	< 0.005	8.20
Architectural Coatings	0.61	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Onsite truck	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Offsite	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.03	0.52	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	143	143	< 0.005	< 0.005	144
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.04	0.03	0.47	0.00	0.00	0.14	0.14	0.00	0.03	0.03	—	135	135	< 0.005	0.01	137
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Average Daily	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	0.01	0.01	0.17	0.00	0.00	0.05	0.05	0.00	0.01	0.01	—	50.2	50.2	< 0.005	< 0.005	50.4
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Worker	< 0.005	< 0.005	0.03	0.00	0.00	0.01	0.01	0.00	< 0.005	< 0.005	—	8.31	8.31	< 0.005	< 0.005	8.34
Vendor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00
Hauling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—	0.00	0.00	0.00	0.00	0.00

4. Operations Emissions Details

4.10. Soil Carbon Accumulation By Vegetation Type

4.10.1. Soil Carbon Accumulation By Vegetation Type - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Vegetation	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e

Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

4.10.2. Above and Belowground Carbon Accumulation by Land Use Type - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Land Use	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

4.10.3. Avoided and Sequestered Emissions by Species - Unmitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Species	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Avoided	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sequestered	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Removed	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Avoided	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sequestered	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Removed	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Avoided	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sequestered	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Removed	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

4.10.4. Soil Carbon Accumulation By Vegetation Type - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Vegetation	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

4.10.5. Above and Belowground Carbon Accumulation by Land Use Type - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Land Use	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

4.10.6. Avoided and Sequestered Emissions by Species - Mitigated

Criteria Pollutants (lb/day for daily, ton/yr for annual) and GHGs (lb/day for daily, MT/yr for annual)

Species	ROG	NOx	CO	SO2	PM10E	PM10D	PM10T	PM2.5E	PM2.5D	PM2.5T	BCO2	NBCO2	CO2T	CH4	N2O	CO2e
Daily, Summer (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Avoided	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sequestered	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Removed	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Daily, Winter (Max)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Avoided	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sequestered	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Removed	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Annual	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Avoided	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sequestered	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Removed	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Subtotal	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

5. Activity Data

5.1. Construction Schedule

Phase Name	Phase Type	Start Date	End Date	Days Per Week	Work Days per Phase	Phase Description
Demolition	Demolition	1/4/2027	3/1/2027	5.00	41.0	—
Site Preparation	Site Preparation	1/4/2027	3/1/2027	5.00	41.0	—
Grading	Grading	2/4/2027	4/29/2027	5.00	61.0	—
Building Construction	Building Construction	4/6/2027	11/2/2027	5.00	151	—
Paving	Paving	5/6/2028	6/30/2028	5.00	40.0	—
Architectural Coating & Finishing	Architectural Coating	9/6/2027	7/7/2028	5.00	220	—

5.2. Off-Road Equipment

5.2.1. Unmitigated

Phase Name	Equipment Type	Fuel Type	Engine Tier	Number per Day	Hours Per Day	Horsepower	Load Factor
Demolition	Rubber Tired Dozers	Diesel	Average	1.00	8.00	367	0.40
Demolition	Crawler Tractors	Diesel	Average	1.00	8.00	300	0.43
Demolition	Excavators	Diesel	Average	1.00	8.00	273	0.38
Demolition	Rubber Tired Loaders	Diesel	Average	1.00	8.00	600	0.36
Demolition	Sweepers/Scrubbers	Diesel	Average	1.00	8.00	36.0	0.46
Site Preparation	Graders	Diesel	Average	1.00	8.00	148	0.41
Site Preparation	Tractors/Loaders/Backhoes	Diesel	Average	1.00	8.00	84.0	0.37

Site Preparation	Rubber Tired Dozers	Diesel	Average	1.00	7.00	367	0.40
Grading	Graders	Diesel	Average	1.00	8.00	148	0.41
Grading	Rubber Tired Dozers	Diesel	Average	1.00	8.00	367	0.40
Grading	Tractors/Loaders/Backhoes	Diesel	Average	2.00	7.00	84.0	0.37
Building Construction	Aerial Lifts	Diesel	Average	4.00	8.00	46.0	0.31
Building Construction	Air Compressors	Diesel	Average	1.00	8.00	49.0	0.48
Building Construction	Bore/Drill Rigs	Diesel	Average	1.00	8.00	83.0	0.50
Building Construction	Cement and Mortar Mixers	Diesel	Average	1.00	8.00	10.0	0.56
Building Construction	Forklifts	Diesel	Average	1.00	8.00	82.0	0.20
Building Construction	Generator Sets	Diesel	Average	1.00	8.00	14.0	0.74
Building Construction	Pumps	Diesel	Average	1.00	8.00	11.0	0.74
Building Construction	Rough Terrain Forklifts	Diesel	Average	1.00	8.00	130	0.40
Building Construction	Sweepers/Scrubbers	Diesel	Average	1.00	8.00	36.0	0.46
Paving	Cement and Mortar Mixers	Diesel	Average	1.00	6.00	10.0	0.56
Paving	Pavers	Diesel	Average	1.00	6.00	81.0	0.42
Paving	Rollers	Diesel	Average	1.00	7.00	36.0	0.38
Paving	Tractors/Loaders/Backhoes	Diesel	Average	1.00	8.00	84.0	0.37
Paving	Paving Equipment	Diesel	Average	1.00	8.00	89.0	0.36
Architectural Coating & Finishing	Air Compressors	Diesel	Average	1.00	6.00	37.0	0.48

5.2.2. Mitigated

Phase Name	Equipment Type	Fuel Type	Engine Tier	Number per Day	Hours Per Day	Horsepower	Load Factor
Demolition	Rubber Tired Dozers	Diesel	Tier 4 Final	1.00	8.00	367	0.40
Demolition	Crawler Tractors	Diesel	Tier 4 Final	1.00	8.00	300	0.43

Demolition	Excavators	Diesel	Tier 4 Final	1.00	8.00	273	0.38
Demolition	Rubber Tired Loaders	Diesel	Tier 4 Final	1.00	8.00	600	0.36
Demolition	Sweepers/Scrubbers	Diesel	Tier 4 Final	1.00	8.00	36.0	0.46
Site Preparation	Graders	Diesel	Tier 4 Final	1.00	8.00	148	0.41
Site Preparation	Tractors/Loaders/Backhoes	Diesel	Tier 4 Final	1.00	8.00	84.0	0.37
Site Preparation	Rubber Tired Dozers	Diesel	Tier 4 Final	1.00	7.00	367	0.40
Grading	Graders	Diesel	Tier 4 Final	1.00	8.00	148	0.41
Grading	Rubber Tired Dozers	Diesel	Tier 4 Final	1.00	8.00	367	0.40
Grading	Tractors/Loaders/Backhoes	Diesel	Tier 4 Final	2.00	7.00	84.0	0.37
Building Construction	Aerial Lifts	Diesel	Tier 4 Final	4.00	8.00	46.0	0.31
Building Construction	Air Compressors	Diesel	Tier 4 Final	1.00	8.00	49.0	0.48
Building Construction	Bore/Drill Rigs	Diesel	Tier 4 Final	1.00	8.00	83.0	0.50
Building Construction	Cement and Mortar Mixers	Diesel	Average	1.00	8.00	10.0	0.56
Building Construction	Forklifts	Diesel	Tier 4 Final	1.00	8.00	82.0	0.20
Building Construction	Generator Sets	Diesel	Average	1.00	8.00	14.0	0.74
Building Construction	Pumps	Diesel	Average	1.00	8.00	11.0	0.74
Building Construction	Rough Terrain Forklifts	Diesel	Tier 4 Final	1.00	8.00	130	0.40
Building Construction	Sweepers/Scrubbers	Diesel	Tier 4 Final	1.00	8.00	36.0	0.46
Paving	Cement and Mortar Mixers	Diesel	Average	1.00	6.00	10.0	0.56
Paving	Pavers	Diesel	Tier 4 Final	1.00	6.00	81.0	0.42
Paving	Rollers	Diesel	Tier 4 Final	1.00	7.00	36.0	0.38
Paving	Tractors/Loaders/Backhoes	Diesel	Tier 4 Final	1.00	8.00	84.0	0.37
Paving	Paving Equipment	Diesel	Tier 4 Final	1.00	8.00	89.0	0.36
Architectural Coating & Finishing	Air Compressors	Diesel	Tier 4 Final	1.00	6.00	37.0	0.48

5.3. Construction Vehicles

5.3.1. Unmitigated

Phase Name	Trip Type	One-Way Trips per Day	Miles per Trip	Vehicle Mix
Demolition	—	—	—	—
Demolition	Worker	12.5	11.7	LDA,LDT1,LDT2
Demolition	Vendor	—	8.40	HHDT,MHDT
Demolition	Hauling	3.93	20.0	HHDT
Demolition	Onsite truck	—	—	HHDT
Site Preparation	—	—	—	—
Site Preparation	Worker	7.50	11.7	LDA,LDT1,LDT2
Site Preparation	Vendor	—	8.40	HHDT,MHDT
Site Preparation	Hauling	0.00	20.0	HHDT
Site Preparation	Onsite truck	—	—	HHDT
Grading	—	—	—	—
Grading	Worker	10.0	11.7	LDA,LDT1,LDT2
Grading	Vendor	—	8.40	HHDT,MHDT
Grading	Hauling	18.6	20.0	HHDT
Grading	Onsite truck	—	—	HHDT
Building Construction	—	—	—	—
Building Construction	Worker	83.7	11.7	LDA,LDT1,LDT2
Building Construction	Vendor	13.7	8.40	HHDT,MHDT
Building Construction	Hauling	0.00	20.0	HHDT
Building Construction	Onsite truck	—	—	HHDT
Paving	—	—	—	—
Paving	Worker	12.5	11.7	LDA,LDT1,LDT2
Paving	Vendor	—	8.40	HHDT,MHDT

Paving	Hauling	0.00	20.0	HHDT
Paving	Onsite truck	—	—	HHDT
Architectural Coating & Finishing	—	—	—	—
Architectural Coating & Finishing	Worker	16.7	11.7	LDA,LDT1,LDT2
Architectural Coating & Finishing	Vendor	—	8.40	HHDT,MHDT
Architectural Coating & Finishing	Hauling	0.00	20.0	HHDT
Architectural Coating & Finishing	Onsite truck	—	—	HHDT

5.3.2. Mitigated

Phase Name	Trip Type	One-Way Trips per Day	Miles per Trip	Vehicle Mix
Demolition	—	—	—	—
Demolition	Worker	12.5	11.7	LDA,LDT1,LDT2
Demolition	Vendor	—	8.40	HHDT,MHDT
Demolition	Hauling	3.93	20.0	HHDT
Demolition	Onsite truck	—	—	HHDT
Site Preparation	—	—	—	—
Site Preparation	Worker	7.50	11.7	LDA,LDT1,LDT2
Site Preparation	Vendor	—	8.40	HHDT,MHDT
Site Preparation	Hauling	0.00	20.0	HHDT
Site Preparation	Onsite truck	—	—	HHDT
Grading	—	—	—	—
Grading	Worker	10.0	11.7	LDA,LDT1,LDT2
Grading	Vendor	—	8.40	HHDT,MHDT
Grading	Hauling	18.6	20.0	HHDT
Grading	Onsite truck	—	—	HHDT
Building Construction	—	—	—	—
Building Construction	Worker	83.7	11.7	LDA,LDT1,LDT2

Building Construction	Vendor	13.7	8.40	HHDT,MHDT
Building Construction	Hauling	0.00	20.0	HHDT
Building Construction	Onsite truck	—	—	HHDT
Paving	—	—	—	—
Paving	Worker	12.5	11.7	LDA,LDT1,LDT2
Paving	Vendor	—	8.40	HHDT,MHDT
Paving	Hauling	0.00	20.0	HHDT
Paving	Onsite truck	—	—	HHDT
Architectural Coating & Finishing	—	—	—	—
Architectural Coating & Finishing	Worker	16.7	11.7	LDA,LDT1,LDT2
Architectural Coating & Finishing	Vendor	—	8.40	HHDT,MHDT
Architectural Coating & Finishing	Hauling	0.00	20.0	HHDT
Architectural Coating & Finishing	Onsite truck	—	—	HHDT

5.4. Vehicles

5.4.1. Construction Vehicle Control Strategies

Control Strategies Applied	PM10 Reduction	PM2.5 Reduction
Water unpaved roads twice daily	55%	55%
Limit vehicle speeds on unpaved roads to 25 mph	44%	44%
Sweep paved roads once per month	9%	9%

5.5. Architectural Coatings

Phase Name	Residential Interior Area Coated (sq ft)	Residential Exterior Area Coated (sq ft)	Non-Residential Interior Area Coated (sq ft)	Non-Residential Exterior Area Coated (sq ft)	Parking Area Coated (sq ft)
Architectural Coating & Finishing	286,700	95,567	0.00	0.00	—

5.6. Dust Mitigation

5.6.1. Construction Earthmoving Activities

Phase Name	Material Imported (cy)	Material Exported (cy)	Acres Graded (acres)	Material Demolished (Building Square Footage)	Acres Paved (acres)
Demolition	0.00	0.00	0.00	14,000	—
Site Preparation	—	—	38.4	0.00	—
Grading	—	9,060	40.0	0.00	—
Paving	0.00	0.00	0.00	0.00	0.00

5.6.2. Construction Earthmoving Control Strategies

Control Strategies Applied	Frequency (per day)	PM10 Reduction	PM2.5 Reduction
Water Exposed Area	2	61%	61%
Water Demolished Area	2	36%	36%

5.7. Construction Paving

Land Use	Area Paved (acres)	% Asphalt
Congregate Care (Assisted Living)	—	0%
Enclosed Parking with Elevator	0.00	100%

5.8. Construction Electricity Consumption and Emissions Factors

kWh per Year and Emission Factor (lb/MWh)

Year	kWh per Year	CO2	CH4	N2O
2027	0.00	204	0.03	< 0.005
2028	0.00	204	0.03	< 0.005

5.18. Vegetation

5.18.1. Land Use Change

5.18.1.1. Unmitigated

Vegetation Land Use Type	Vegetation Soil Type	Initial Acres	Final Acres

5.18.1.2. Mitigated

Vegetation Land Use Type	Vegetation Soil Type	Initial Acres	Final Acres

5.18.1. Biomass Cover Type

5.18.1.1. Unmitigated

Biomass Cover Type	Initial Acres	Final Acres

5.18.1.2. Mitigated

Biomass Cover Type	Initial Acres	Final Acres

5.18.2. Sequestration

5.18.2.1. Unmitigated

Tree Type	Number	Electricity Saved (kWh/year)	Natural Gas Saved (btu/year)

5.18.2.2. Mitigated

Tree Type	Number	Electricity Saved (kWh/year)	Natural Gas Saved (btu/year)

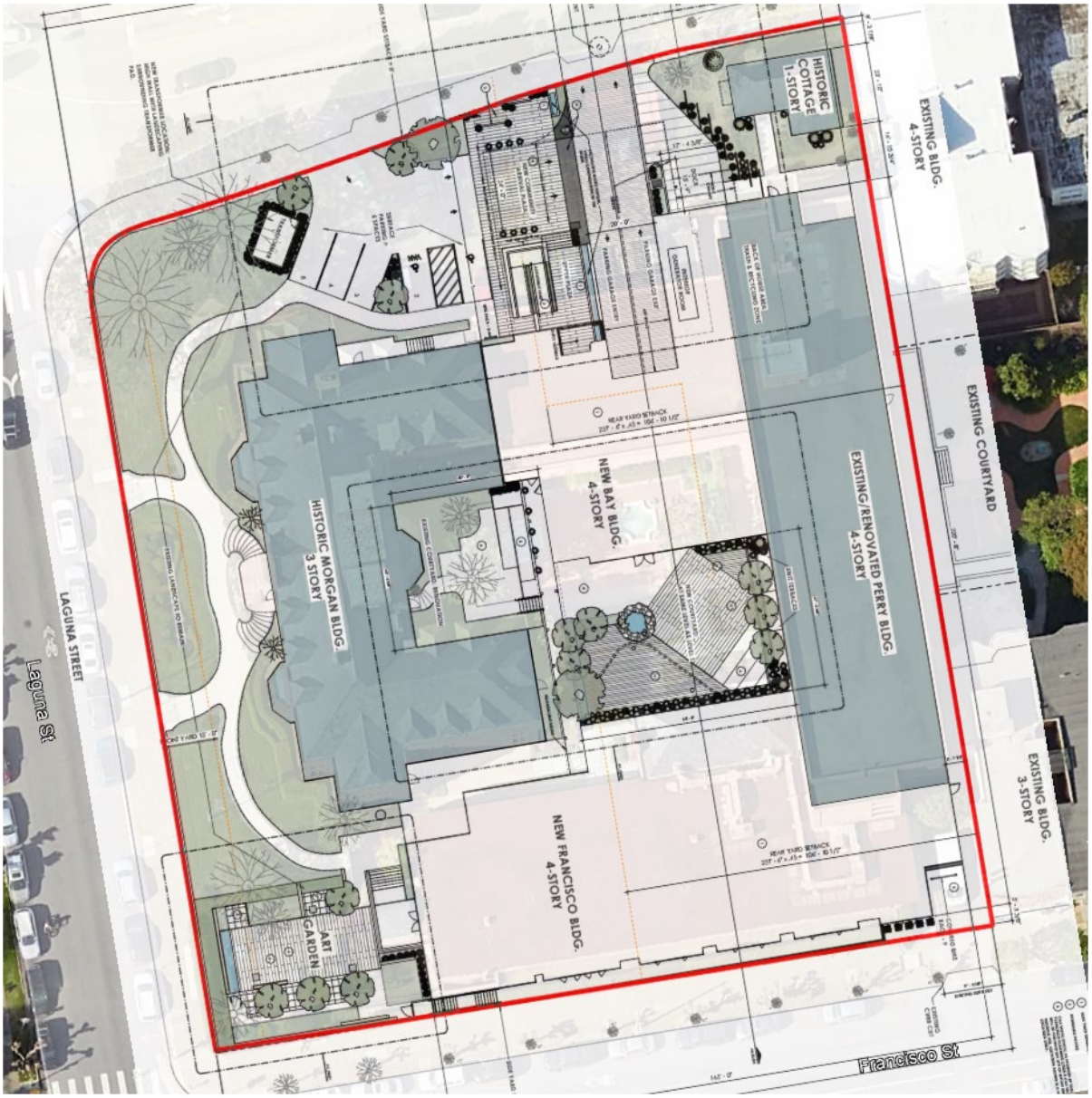
8. User Changes to Default Data

Screen	Justification
Land Use	Based on Project Description.
Construction: Construction Phases	Based on Construction Phasing provided by Applicant.
Construction: Off-Road Equipment	Based on construction equipment list provided by Project Applicant.
Construction: Dust From Material Movement	The proposed project would require approximately 9,060 cubic yards off-haul.

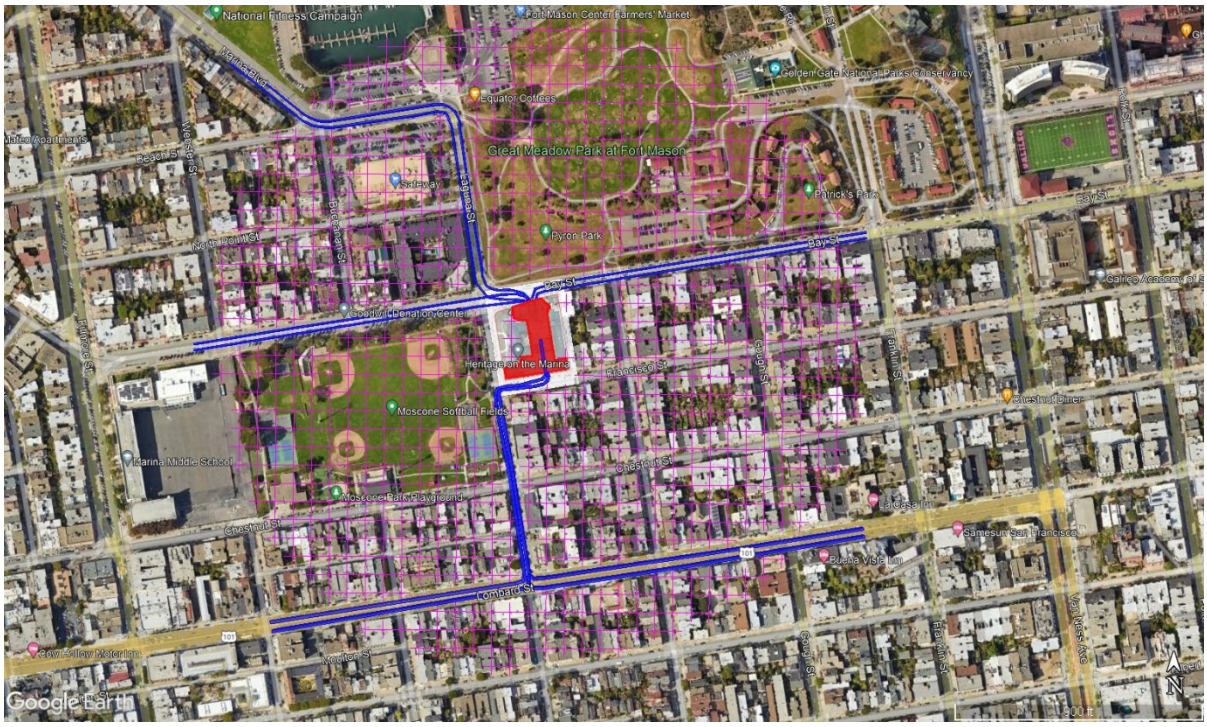
ATTACHMENT B

HRA MODEL SNAPSHOTS

Project Layout



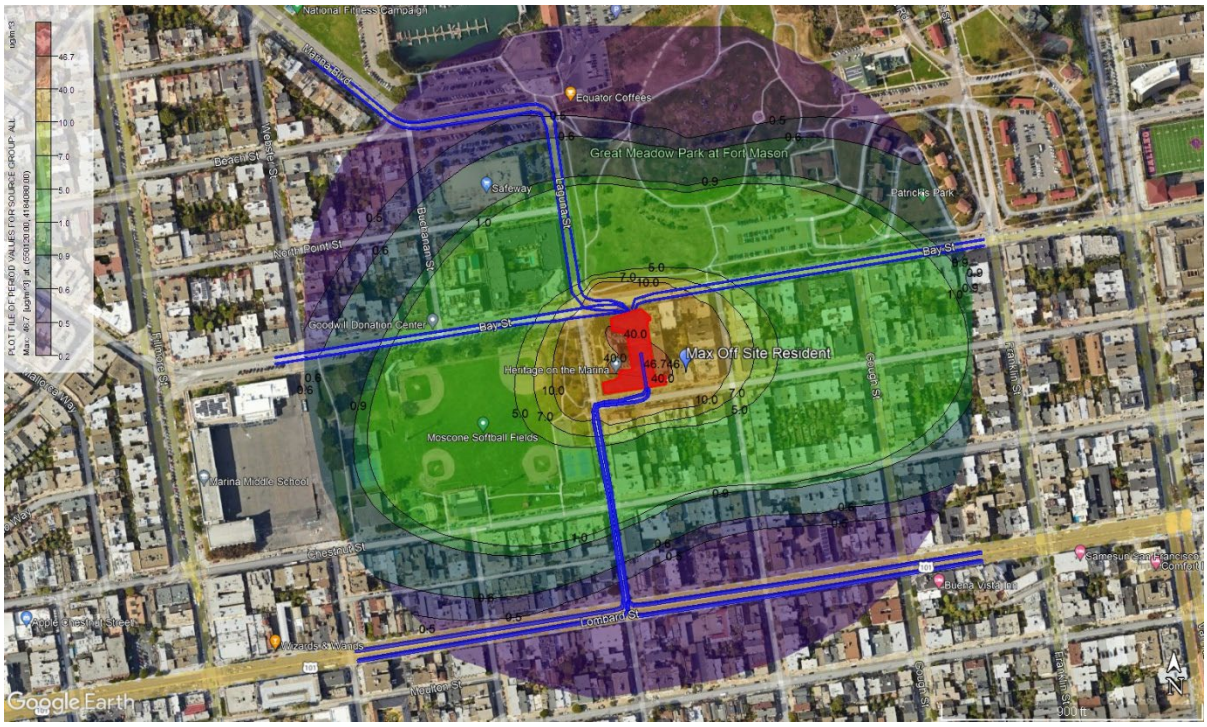
Project Grid and Sources



Uncontrolled Cancer Risk - On-site Residential Receptor



Uncontrolled Cancer Risk - Off-site Residential Receptor



Uncontrolled Cancer Risk - Off-site Childcare Receptor



Uncontrolled Cancer Risk - Off-site School Receptor



Uncontrolled Cancer Risk - On-site and Off-site Worker Receptors



Uncontrolled $PM_{2.5}$ Concentration



Uncontrolled Chronic Hazard Index - On-site Residential Receptor



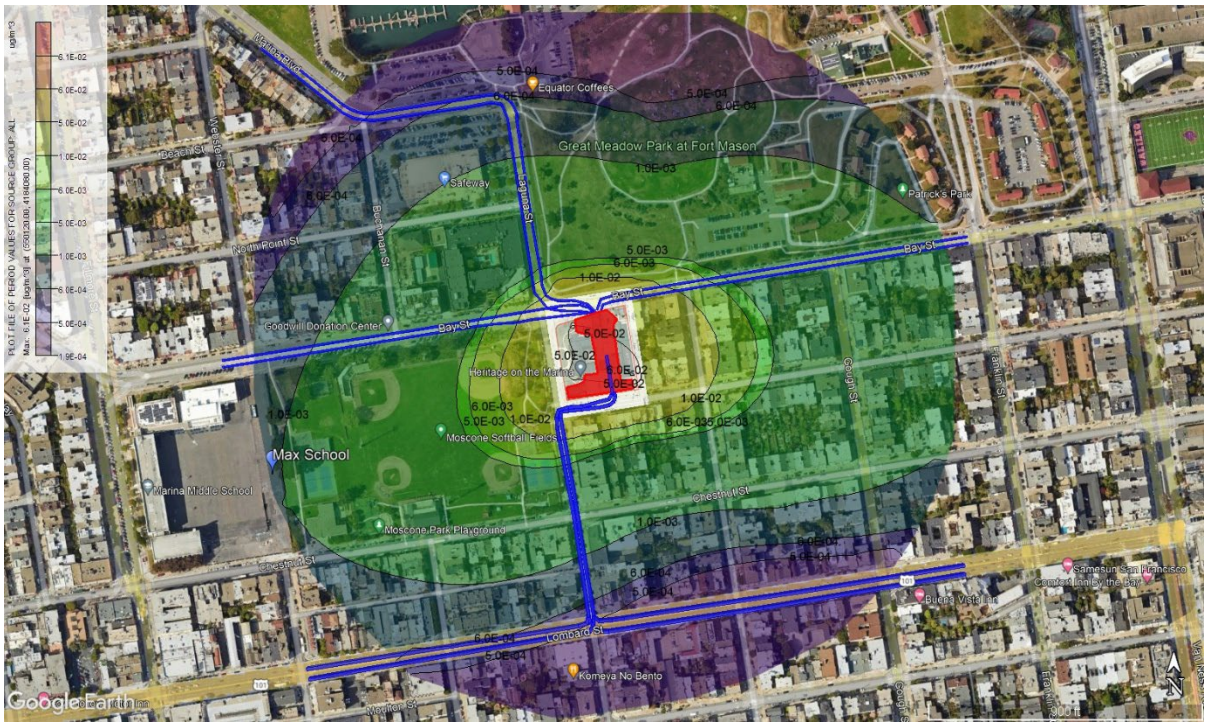
Uncontrolled Chronic Hazard Index - Off-site Residential Receptor



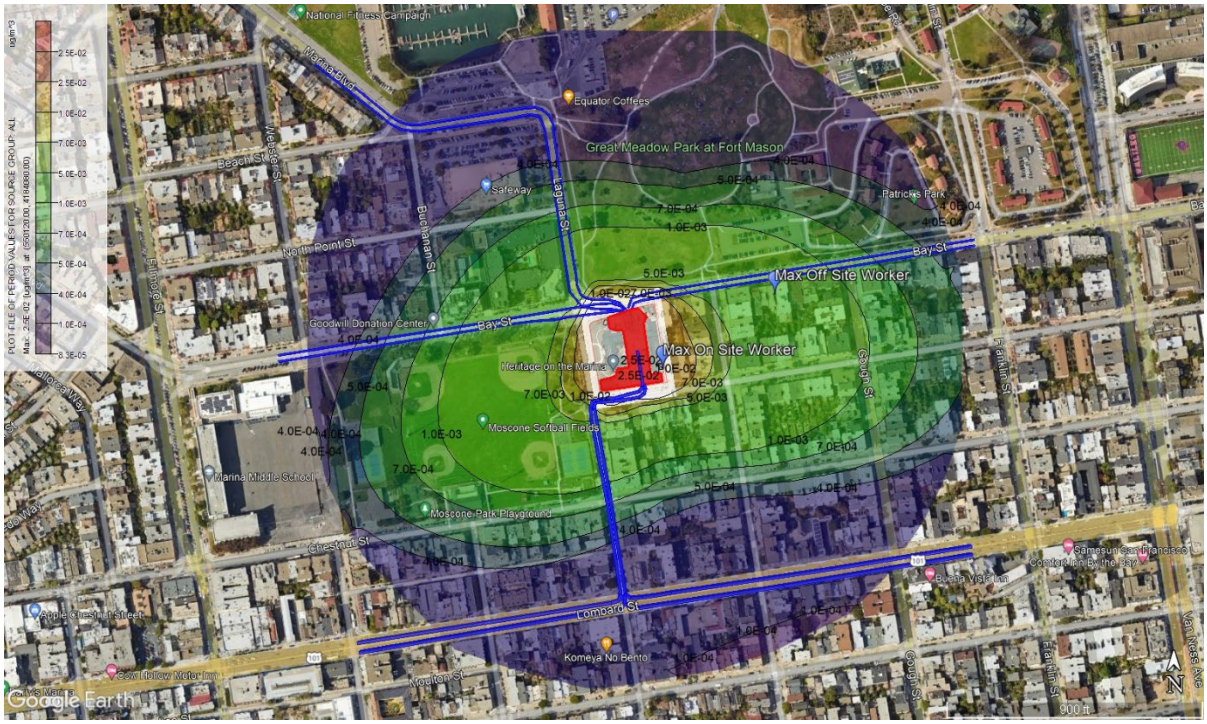
Uncontrolled Chronic Hazard Index - Off-site Childcare Receptor



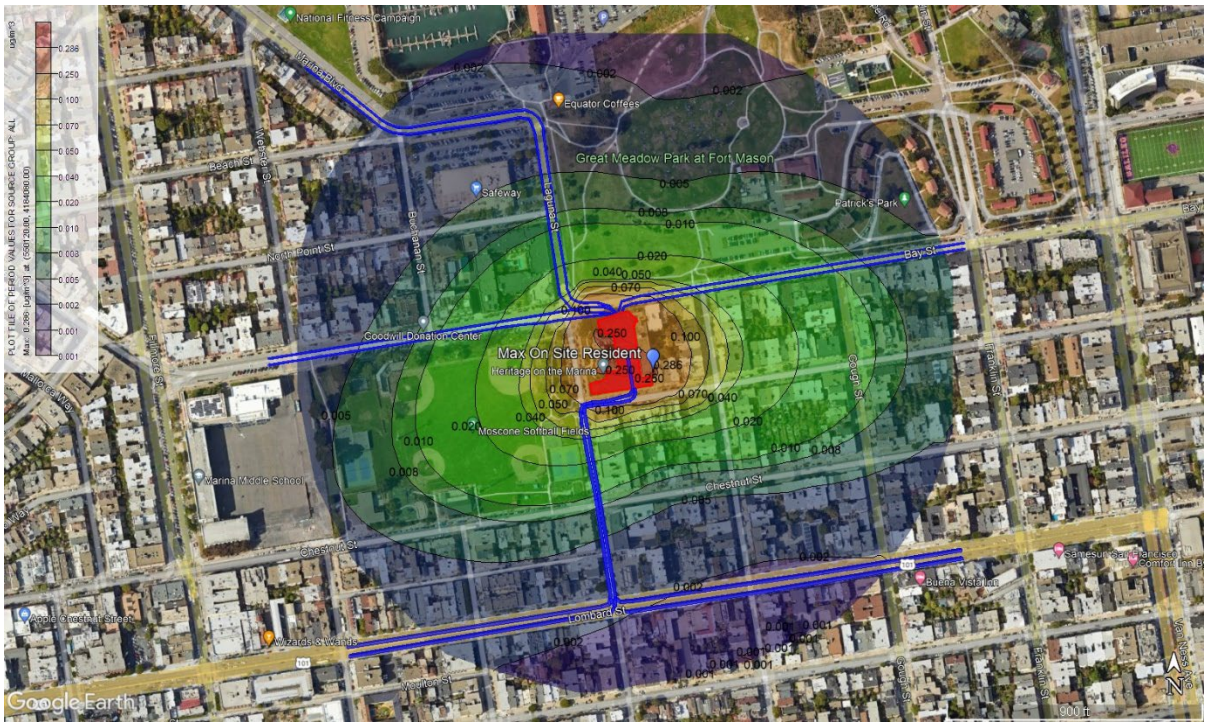
Uncontrolled Chronic Hazard Index - Off-site School Receptor



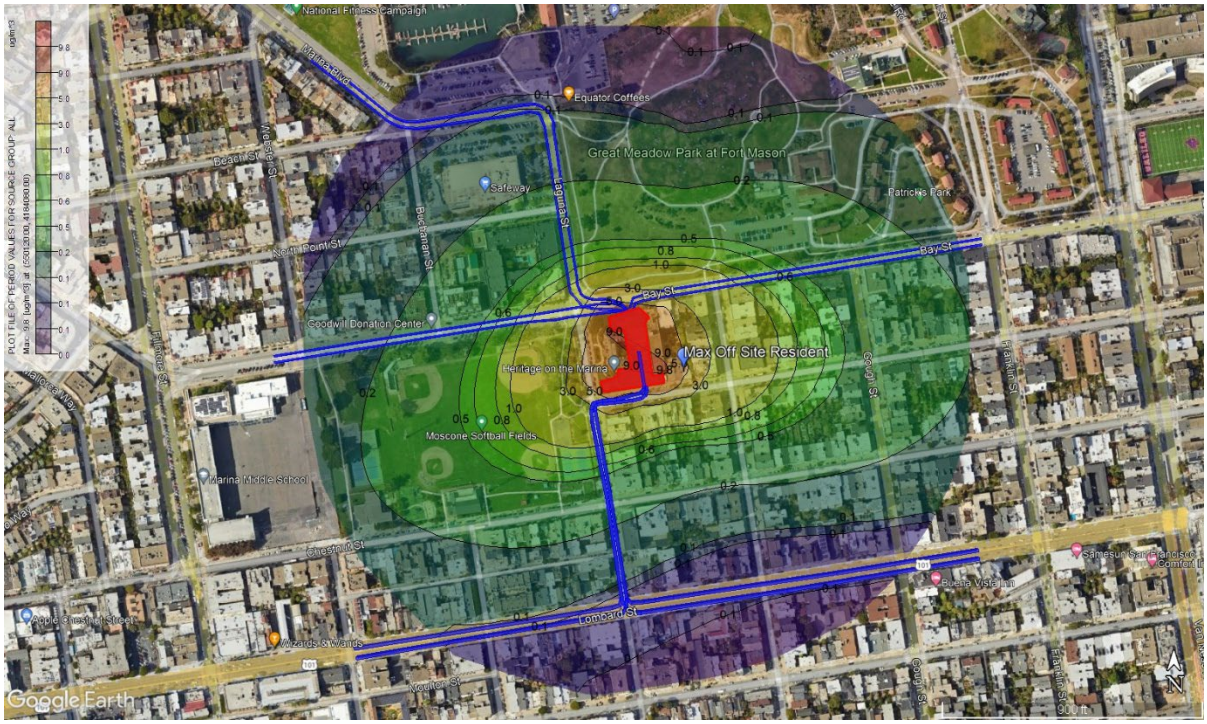
Uncontrolled Chronic Hazard Index - On-site and Off-site Worker Receptors



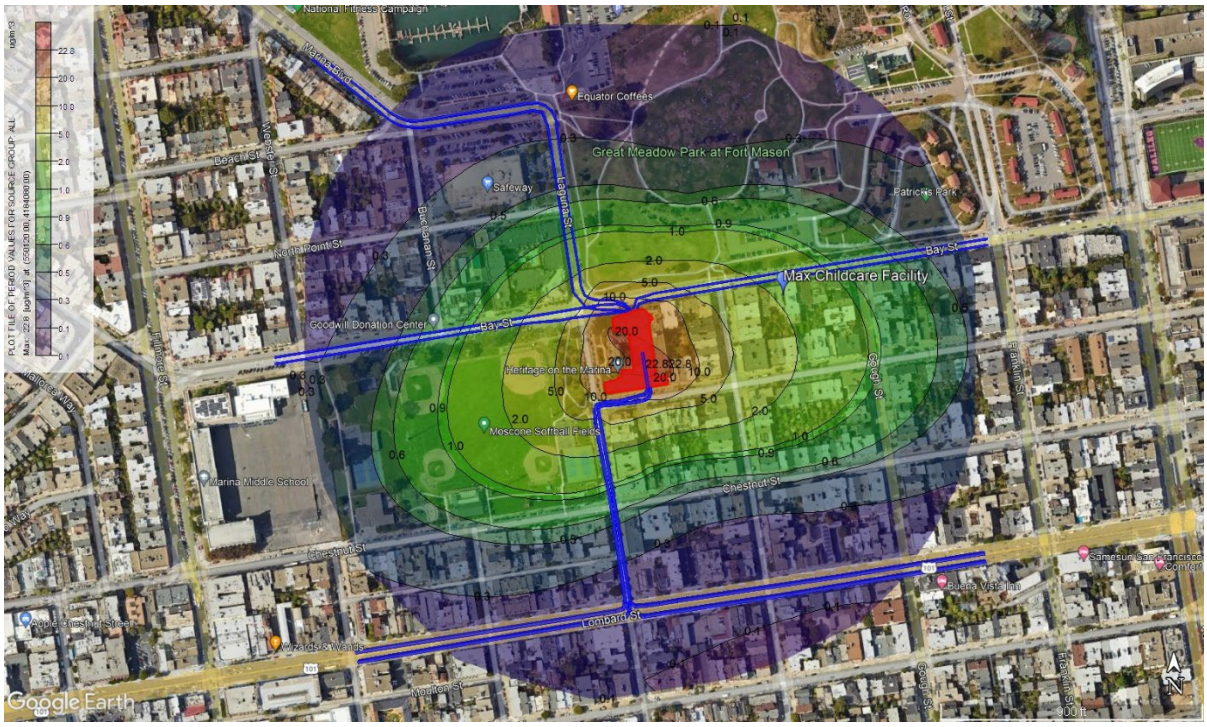
Controlled Cancer Risk - On-site Residential Receptor



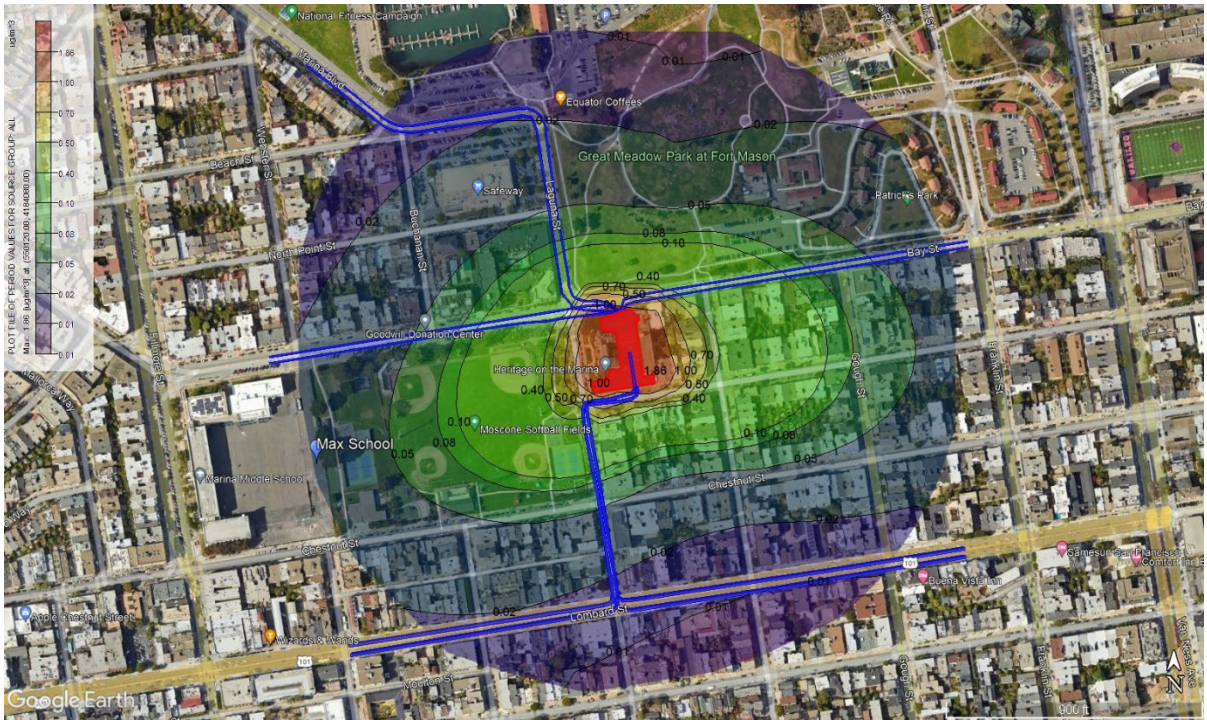
Controlled Cancer Risk - Off-site Residential Receptor



Controlled Cancer Risk - Off-site Childcare Receptor



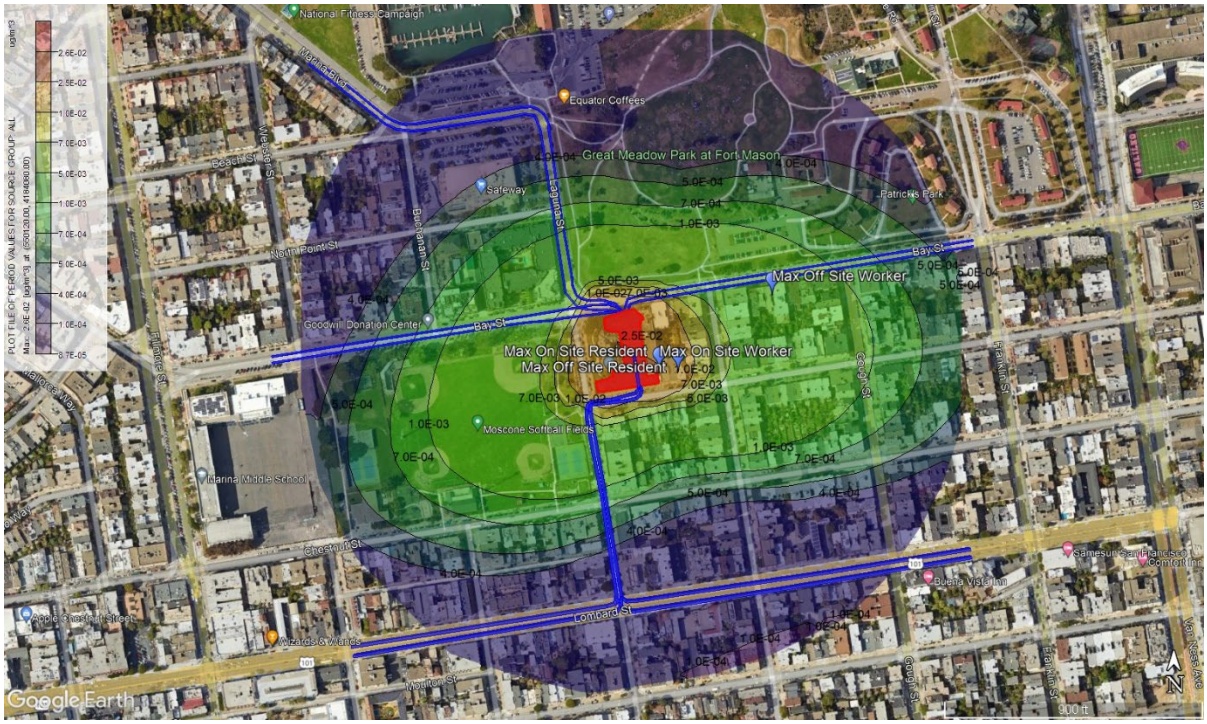
Controlled Cancer Risk - Off-site School Receptor



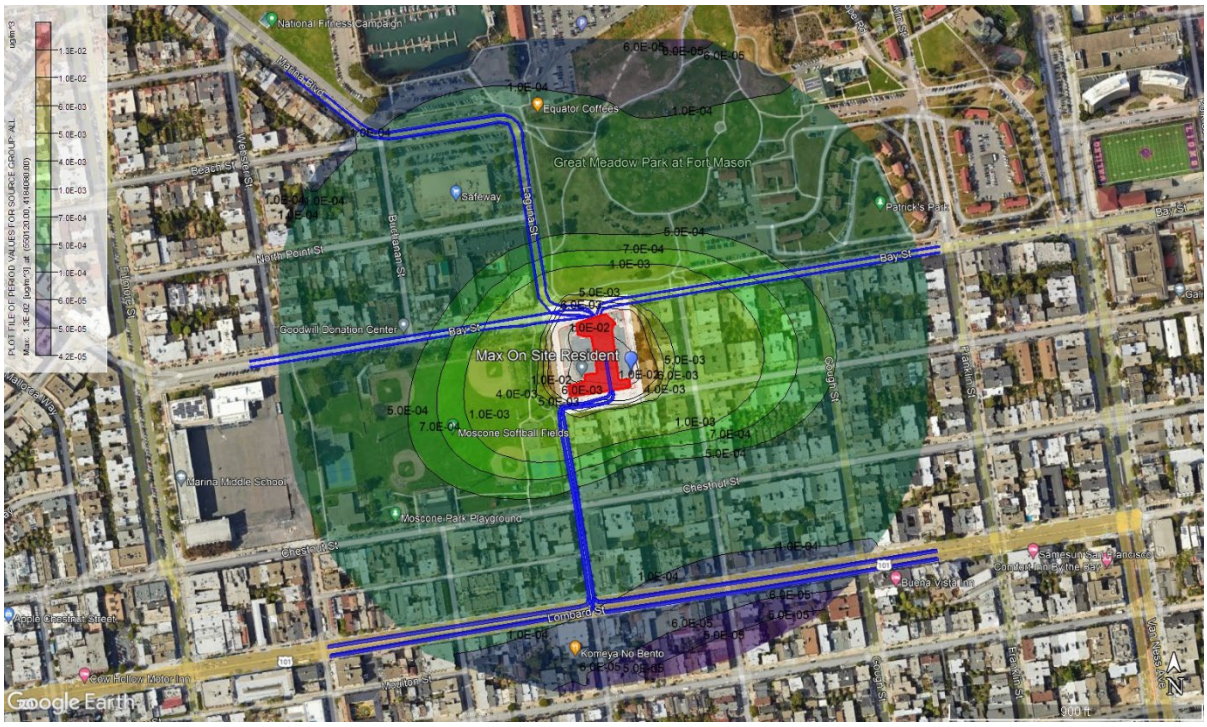
Controlled Cancer Risk - On-site and Off-site Worker Receptors



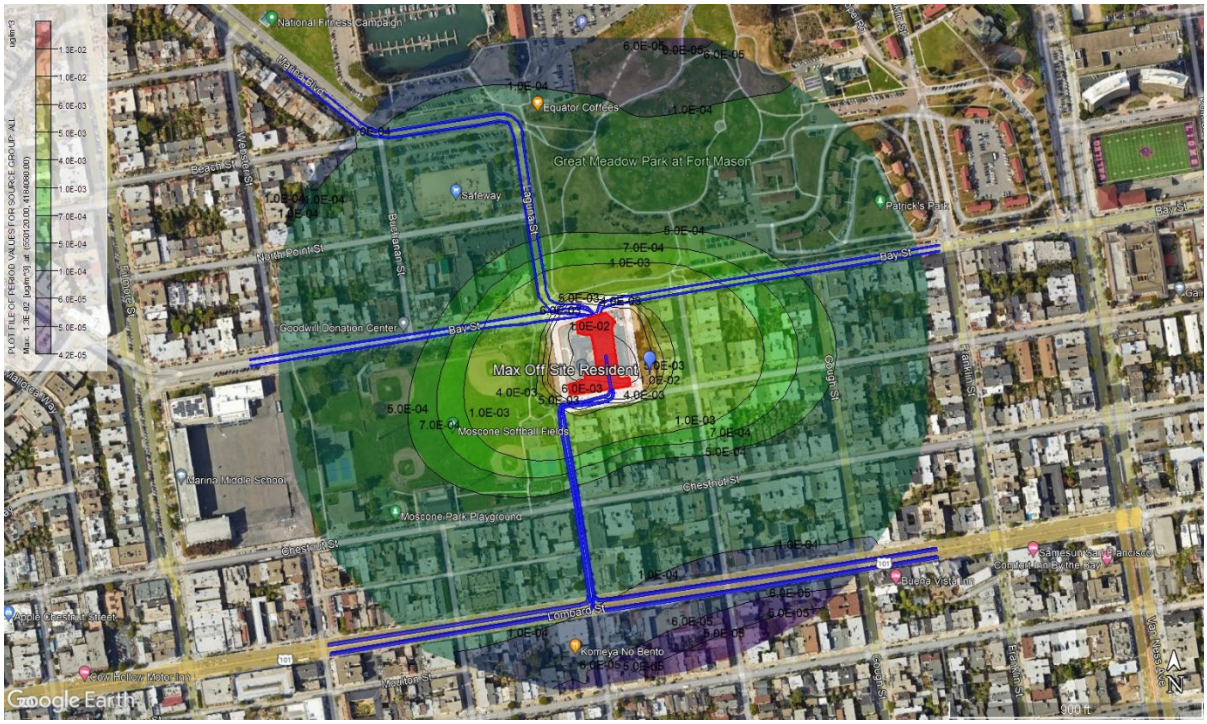
Controlled $PM_{2.5}$ Concentration



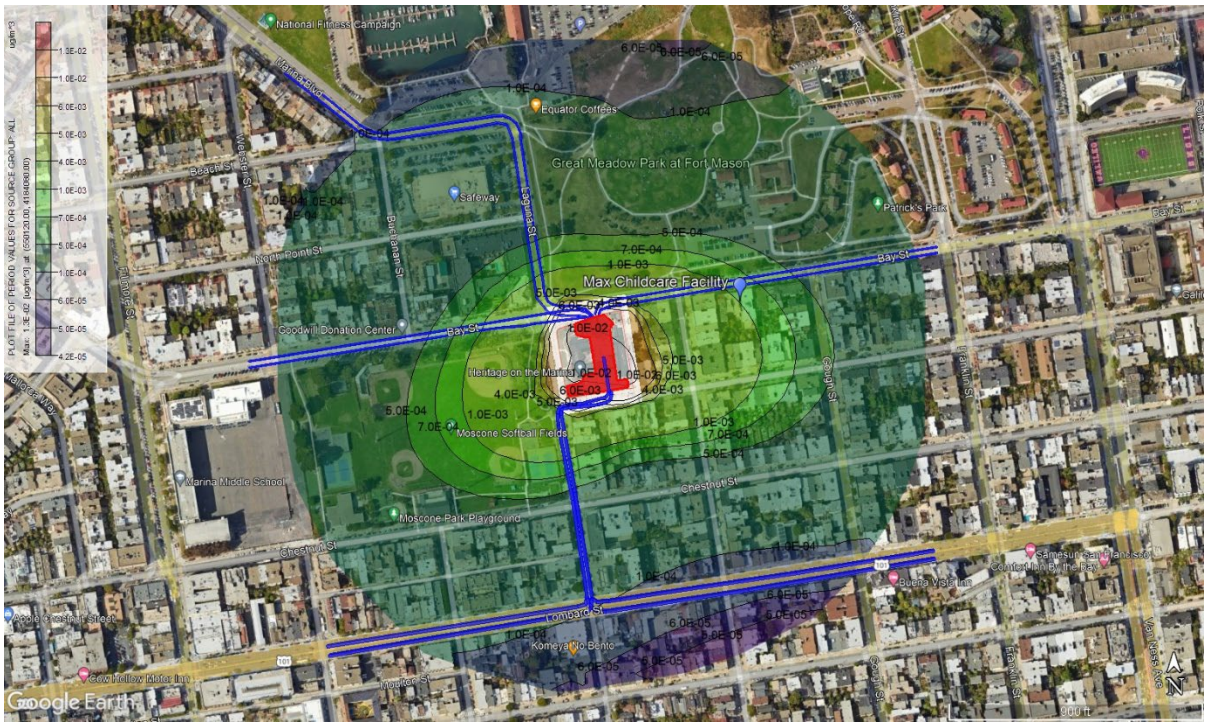
Controlled Chronic Hazard Index - On-site Residential Receptor



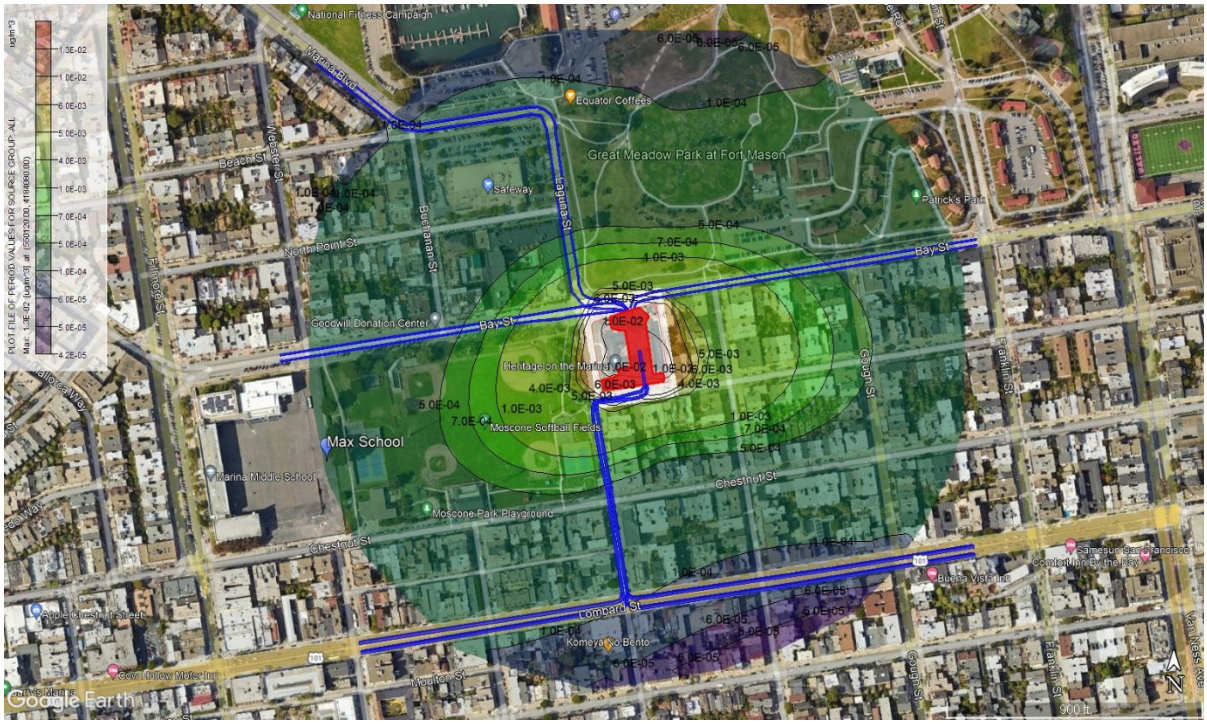
Controlled Chronic Hazard Index - Off-site Residential Receptor



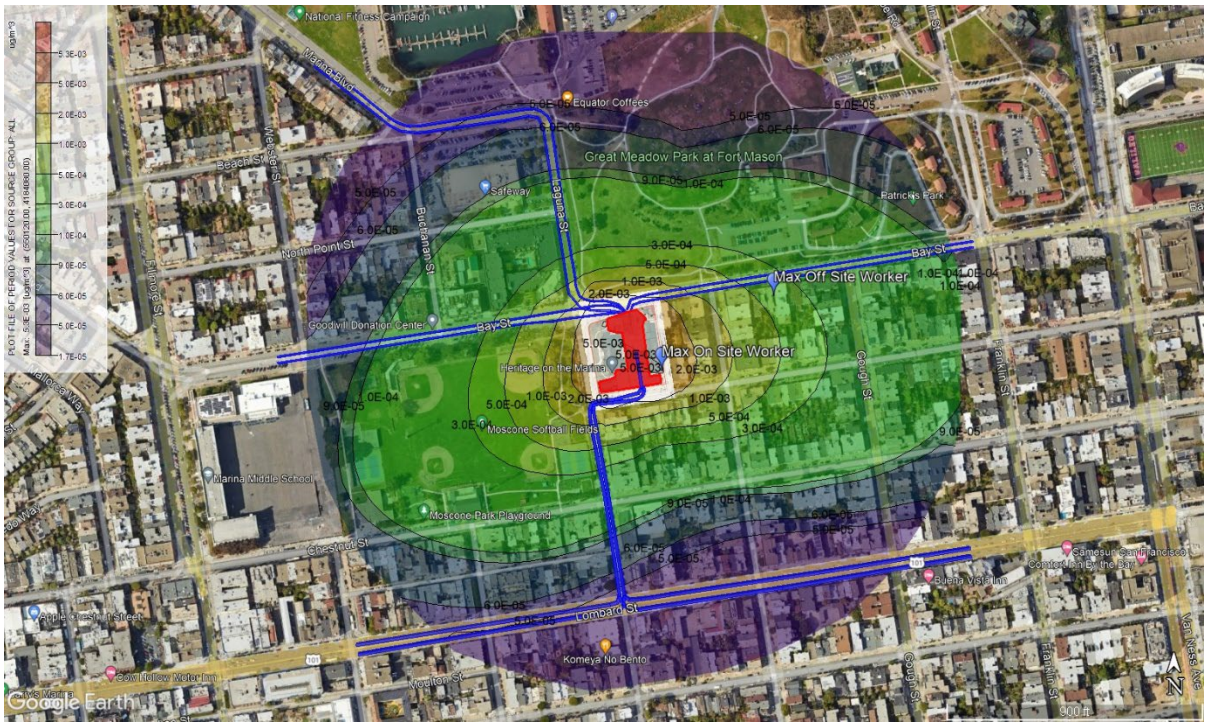
Controlled Chronic Hazard Index - Off-site Childcare Receptor



Controlled Chronic Hazard Index - Off-site School Receptor



Controlled Chronic Hazard Index - On-site and Off-site Worker Receptors



				In a Million					PM 2.5					Chronic HI				
Type	X	Y	Code	Average	T2L3	T4i	T4f	T4F-ElectGen	Average	T2L3	T4i	T4f	T4F-ElectGen	Average	T2L3	T4i	T4f	T4F-ElectGen
On Site Resident	550120	4184080	550120_4184080	1.37	0.40	0.68	0.29	0.24	0.126	0.037	0.063	0.022	0.029	0.061	0.018	0.030	0.013	0.004
Off Site Resident	550140	4184080	550140_4184080	26.31	7.63	13.12	5.49	4.58	0.071	0.021	0.035	0.015	0.012	0.034	0.010	0.017	0.007	0.002
Childcare	550240	4184160	550240_4184160	8.44	2.45	4.21	1.76	1.47	0.023	0.007	0.012	0.005	0.002	0.005	0.001	0.002	0.001	0.001
School	549760	4183980	549760_4183980	0.1381	0.04	0.07	0.03	0.02	0.005	0.001	0.002	9.81E-04	8.18E-04	9.40E-04	2.73E-04	4.69E-04	1.96E-04	6.82E-05
Worker On Site	550120	4184080	550120_4184080	0.69	0.20	0.35	0.14	0.12	0.126	0.037	0.063	0.026	0.029	0.025	0.007	0.013	0.005	0.004
Worker Off Site	550240	4184160	550240_4184160	0.053	0.015	0.027	0.011	0.009	0.010	0.003	0.005	0.002	0.002	0.002	0.001	0.001	0.000	0.000

Receptor Type	Age Group	Daily Breathing Rate (L/kg day or L/kg 8 hours)	Fraction of Time at Home	Exposure Frequency (days per year)	Exposure Duration (years)	Model Adjustment Factor	Age Sensitivity Factor	Conversion Factor (m^3/L)	Averaging Time (days)
Offsite Resident	Third Trimester	240	1	350	0.25	1	10	0.001	25,550
	Age 0-2 Years	1200	1	350	1	1	10	0.001	25,550
	Age 2-16 Years	520	1	350	1	1	3	0.001	25,550
	16-30	240	0.73	350	0	1	1	0.001	25,550
Offsite Childcare	Age 0-2 Years	1,200	1	350	1	4.2	10	0.001	25,550
	Age 2-9 Years	640	1	350	1	4.2	3	0.001	25,550
Offsite School	Age 2-16 Years	520	1	350	1	4.2	3	0.001	25,550

Chemical/TAC	EMFAC Gasoline TOG Speciation (% TOG)	Inhalation Cancer Potency Factors (mg/kg-day)$^{-1}$	Weighted Inhalation Cancer Potency Factor
Diesel Exhaust (Diesel Vehicles)			
Diesel Particulate Matter	n/a	1.1	n/a
TOG Exhaust (Gasoline Vehicles)			
Acetaldehyde	0.28%	1.00E-02	2.80E-05
Benzene	2.47%	1.00E-01	2.47E-03
1,3-Butadiene	0.55%	6.00E-01	3.30E-03
Ethylbenzene	1.05%	8.70E-03	9.14E-05
Formaldehyde	1.58%	2.10E-02	3.32E-04
Naphthalene	0.05%	1.20E-01	6.00E-05
Weighted Total Toxicity Factor	—	—	6.28E-03
TOG Evaporative (Gasoline Vehicles)			
Benzene	0.36%	0.1	3.60E-04
Ethylbenzene	0.12%	8.70E-03	1.04E-05
Weighted Total Toxicity Factor	—	—	3.70E-04

Construction

Project Area	Years	Days	$PM10_{Exhaust}$	$PM10_{Total}$
Tons/Yr*	2027	233	6.44E-02	0.086
	2028	260	3.96E-03	0.003
		0	0.00E+00	0.00E+00

*CalEEMod

Project Area	Years	$PM10_{Exhaust}$	$PM10_{Total}$
Lbs/Yr	2027	128.80	171.40
	2028	7.93	5.19
		0.00	0.00
	Total	136.73	176.59

Project Area	Years	$PM10_{Exhaust}$	$PM10_{Total}$
g/sec	2027	2.90E-03	3.86E-03
	2028	1.60E-04	1.05E-04
		0.00E+00	0.00E+00

HHDT Idling

	Years	Days	Trips	g/Trip*
Vendor	2027	233	30.5	0.0408
	2028	260	0	
		1	0	
	Years	Days	Trips	
Hauling	2027	233	22.5	
	2028	260	0	
		1	0	

*CalEEmod

Vendor	Years	$PM10_{Exhaust}$	$PM10_{Total}$
Lbs/Yr	2027	0.0027	0.0027
	2028	0.0000	0.0000
		0.0000	0.0000
Hauling	Years	$PM10_{Exhaust}$	$PM10_{Total}$
Lbs/Yr	2027	0.0020	0.0020
	2028	0.0000	0.0000
		0.0000	0.0000
	Total	0.0048	0.0048

Vendor	Years	$PM10_{Exhaust}$	$PM10_{Total}$
g/sec	2027	6.17E-08	6.17E-08
	2028	0.00E+00	0.00E+00
		0.00E+00	0.00E+00
Hauling	Years	$PM10_{Exhaust}$	$PM10_{Total}$
g/sec	2027	4.56E-08	4.56E-08
	2028	0.00E+00	0.00E+00
		0.00E+00	0.00E+00

Vendor & Hualing	Years	$PM10_{Exhaust}$	$PM10_{Total}$
g/sec	2027	1.07E-07	1.07E-07
	2028	0.00E+00	0.00E+00
		0.00E+00	0.00E+00

HHDT Travel

				g/VMT*		
	Years	Days	Trips	PMBW	PMTW	RUNEX
Vendor	2027	233	30.5	0.097	0.034	0.026
	2028	260	0			
		1	0			
	Years	Days	Trips	PMBW	PMTW	RUNEX
Hauling	2027	233	22.5	0.097	0.034	0.026
	2028	260	0			
		1	0			

*EMFAC 2021

Vendor	Years	$PM10_{Exhaust}$	$PM10_{Total}$
Lbs/Yr/mi	2027	0.0017	0.0106
	2028	0.0000	0.0000
		0.0000	0.0000
Hauling	Years	$PM10_{Exhaust}$	$PM10_{Total}$
Lbs/Yr/mi	2027	0.0013	0.0078
	2028	0.0000	0.0000
		0.0000	0.0000

Vendor & Hualing	Years	$PM10_{Exhaust}$	$PM10_{Total}$
Lbs/Yr/mi	2027	0.0030	0.0183
	2028	0.0000	0.0000
	0	0.0000	0.0000

Vendor	Years	$PM10_{Exhaust}$	$PM10_{Total}$
g/sec/mi	2027	3.91E-08	2.38E-07
	2028	0.00E+00	0.00E+00
		0.00E+00	0.00E+00
Hauling	Years	$PM10_{Exhaust}$	$PM10_{Total}$
g/sec/mi	2027	2.88E-08	1.76E-07
	2028	0.00E+00	0.00E+00
		0.00E+00	0.00E+00

Vendor & H	Years	$PM10_{Exhaus}$	$PM10_{Total}$
g/sec/mi	2027	6.79E-08	4.13E-07
	2028	0.00E+00	0.00E+00
		0.00E+00	0.00E+00

	Distance (m)						
Years	TRK1_OUT	TRK2_IN	TRK3_IN	TRK4_OUT	TRK_OUT	TRK_IN	Onsite
2027	289.8	298.2	373.7	385.4	263.2	266.3	38.7
2028							

*1609m = 1 mile

$PM10_{Exhaust}$ / Modeling Source (Lbs/Yr)						
TRK1_OUT	TRK2_IN	TRK3_IN	TRK4_OUT	TRK_OUT	TRK_IN	Onsite
1.09E-03	1.12E-03	1.40E-03	1.44E-03	9.86E-04	9.97E-04	1.45E-04
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

$PM10_{Total}$ / Modeling Source (Lbs/Yr)						
TRK1_OUT	TRK2_IN	TRK3_IN	TRK4_OUT	TRK_OUT	TRK_IN	Onsite
6.61E-03	6.80E-03	8.52E-03	8.79E-03	6.00E-03	6.07E-03	8.83E-04
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Worker Travel

				g/VMT*		
	Years	Days	Trips	LDA	LDT1	LDT2
TOG_RUNEX	2027	233	113.7	0.025	0.099	0.080
	2028	260	29.2			
		1	0			
	Years	Days	Trips	LDA	LDT1	LDT2
TOG_RUNLS	2027	233	113.7	0.225	0.270	0.190
	2028	260	29.2			
		1	0			
	Years	Days	Trips	LDA	LDT1	LDT2
PM10_PMTW	2027	233	113.7	0.008	0.008	0.008
	2028	260	29.2			
		1	0			
	Years	Days	Trips	LDA	LDT1	LDT2
PM10_PMBW	2027	233	113.7	0.007	0.008	0.008
	2028	260	29.2			
		1	0			

*EMFAC 2021

RUNEX	Years	TOG_{Total}
Lbs/Yr/mi	2027	0.0510
	2028	0.0131
		0.0000
RUNLS	Years	TOG_{Total}
Lbs/Yr/mi	2027	0.1717
	2028	0.0441
		0.0000
PMTW	Years	$PM10_{Total}$
Lbs/Yr/mi	2027	0.0060
	2028	0.0015
		0.0000
PMBW	Years	$PM10_{Total}$
Lbs/Yr/mi	2027	0.0059
	2028	0.0015
		0.0000

RUNEX	Years	TOG_{Total}
g/sec/mi	2027	1.15E-06
	2028	2.65E-07
		0.00E+00
RUNLS	Years	TOG_{Total}
g/sec/mi	2027	3.87E-06
	2028	8.91E-07
		0.00E+00
PMTW	Years	$PM10_{Total}$
g/sec/mi	2027	1.36E-07
	2028	3.12E-08
		0.00E+00
PMBW	Years	$PM10_{Total}$
g/sec/mi	2027	1.33E-07
	2028	3.06E-08
		0.00E+00

		Distance (m)						
	Years	VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT	
TOG_RUNEX	2027	517.8	519.2	367.6	374.3	380.4	373.3	
	2028							
	Years	VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT	
TOG_RUNLS	2027	517.8	519.2	367.6	374.3	380.4	373.3	
	2028							

TOG_{Total} / Modeling Source (Lbs/Yr)					
VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
3.28E-02	3.29E-02	2.33E-02	2.37E-02	2.41E-02	2.37E-02
8.45E-03	8.47E-03	6.00E-03	6.11E-03	6.20E-03	6.09E-03
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
1.10E-01	1.11E-01	7.84E-02	7.99E-02	8.12E-02	7.97E-02
2.84E-02	2.85E-02	2.02E-02	2.05E-02	2.09E-02	2.05E-02
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

$PM10_{Total}$ / Modeling Source (Lbs/Yr)					
VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
7.67E-03	7.69E-03	5.44E-03	5.54E-03	5.63E-03	5.53E-03
1.97E-03	1.98E-03	1.40E-03	1.43E-03	1.45E-03	1.42E-03
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

	Pollutants	VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
TOG_RUNEX 2027	Acetaldehyde	9.20E-05	9.22E-05	6.53E-05	6.65E-05	6.76E-05	6.63E-05
	Benzene	8.11E-04	8.13E-04	5.76E-04	5.86E-04	5.96E-04	5.85E-04
	1,3-Butadiene	1.81E-04	1.81E-04	1.28E-04	1.31E-04	1.33E-04	1.30E-04
	Ethylbenzene	3.45E-04	3.46E-04	2.45E-04	2.49E-04	2.53E-04	2.49E-04
	Formaldehyde	5.19E-04	5.20E-04	3.68E-04	3.75E-04	3.81E-04	3.74E-04
	Naphthalene	1.64E-05	1.65E-05	1.17E-05	1.19E-05	1.21E-05	1.18E-05
TOG_RUNLS 2027	Benzene	3.98E-04	3.99E-04	2.82E-04	2.88E-04	2.92E-04	2.87E-04
	Ethylbenzene	1.33E-04	1.33E-04	9.41E-05	9.58E-05	9.74E-05	9.56E-05

2027 Total Lbs/Yr						
Pollutants	VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
Acetaldehyde	9.20E-05	9.22E-05	6.53E-05	6.65E-05	6.76E-05	6.63E-05
Benzene	1.21E-03	1.21E-03	8.58E-04	8.74E-04	8.88E-04	8.72E-04
1,3-Butadiene	1.81E-04	1.81E-04	1.28E-04	1.31E-04	1.33E-04	1.30E-04
Ethylbenzene	4.77E-04	4.79E-04	3.39E-04	3.45E-04	3.51E-04	3.44E-04
Formaldehyde	5.19E-04	5.20E-04	3.68E-04	3.75E-04	3.81E-04	3.74E-04
Naphthalene	1.64E-05	1.65E-05	1.17E-05	1.19E-05	1.21E-05	1.18E-05

All Total Lbs/Yr						
Pollutants	VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
Acetaldehyde	1.16E-04	1.16E-04	8.21E-05	8.36E-05	8.49E-05	8.33E-05
Benzene	1.45E-03	1.45E-03	1.03E-03	1.05E-03	1.06E-03	1.04E-03
1,3-Butadiene	2.27E-04	2.28E-04	1.61E-04	1.64E-04	1.67E-04	1.64E-04
Ethylbenzene	5.76E-04	5.78E-04	4.09E-04	4.17E-04	4.23E-04	4.15E-04
Formaldehyde	6.52E-04	6.54E-04	4.63E-04	4.72E-04	4.79E-04	4.70E-04
Naphthalene	2.06E-05	2.07E-05	1.47E-05	1.49E-05	1.52E-05	1.49E-05

	Pollutants	VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
TOG_RUNEX 2028	Acetaldehyde	2.36E-05	2.37E-05	1.68E-05	1.71E-05	1.74E-05	1.70E-05
	Benzene	2.09E-04	2.09E-04	1.48E-04	1.51E-04	1.53E-04	1.50E-04
	1,3-Butadiene	4.65E-05	4.66E-05	3.30E-05	3.36E-05	3.41E-05	3.35E-05
	Ethylbenzene	8.87E-05	8.89E-05	6.30E-05	6.41E-05	6.51E-05	6.39E-05
	Formaldehyde	1.33E-04	1.34E-04	9.47E-05	9.65E-05	9.80E-05	9.62E-05
	Naphthalene	4.22E-06	4.23E-06	3.00E-06	3.05E-06	3.10E-06	3.04E-06
TOG_RUNLS 2028	Benzene	3.04E-05	3.05E-05	2.16E-05	2.20E-05	2.23E-05	2.19E-05
	Ethylbenzene	1.01E-05	1.02E-05	7.20E-06	7.33E-06	7.45E-06	7.31E-06

2028 Total Lbs/Yr						
Pollutants	VEHN_IN	VEHN_OUT	VEHW_IN	VEHW_OUT	VWHE_IN	VEHE_OUT
Acetaldehyde	2.36E-05	2.37E-05	1.68E-05	1.71E-05	1.74E-05	1.70E-05
Benzene	2.39E-04	2.40E-04	1.70E-04	1.73E-04	1.76E-04	1.72E-04
1,3-Butadiene	4.65E-05	4.66E-05	3.30E-05	3.36E-05	3.41E-05	3.35E-05
Ethylbenzene	9.88E-05	9.91E-05	7.02E-05	7.14E-05	7.26E-05	7.12E-05
Formaldehyde	1.33E-04	1.34E-04	9.47E-05	9.65E-05	9.80E-05	9.62E-05
Naphthalene	4.22E-06	4.23E-06	3.00E-06	3.05E-06	3.10E-06	3.04E-06

Emergency Generator

HP	EF (g/BHP/Hr)*	Hrs /Yr	g/yr	g/sec
131	0.01	50	65.5	0.00018194

*CARB Off Road Compression - Ignition Diesel Engine Standards

Source: EMFAC2021 (v1.0.2) Emission Rates
Region Type: County
Region: San Francisco
Calendar Year: 2027
Season: Annual
Vehicle Classification: EMFAC2007 Categories
Units: miles/day for CVMT and EVMT, trips/day for Trips, g/mile for RUNEX, PMBW and PMTW, g/trip for STREX, HOTSOAK and RUNLOSS, g/vehicle/day for IDLEX and DIURN. PHEV calculated based on total VMT.

Region	Calendar Yr	Vehicle Cat	Model Year	Speed	Fuel	Population	Total VMT	CVMT	EVMT	Trips	NOx_RUNEX	NOx_IDLEX	NOx_STREX	PM2.5_RUNEX	PM2.5_IDLEX	PM2.5_STREX	PM2.5_PMTW	PM2.5_PMBW	PM10_RUNEX	PM10_IDLEX	PM10_STREX	PM10_PMTW	PM10_PMBW	CO2_RUNEX	CO2_IDLEX	CO2_STREX	CH4_RUNEX	CH4_IDLEX	CH4_STREX	N2O_RUNEX	N2O_IDLEX	N2O_STREX	ROG_RUNEX	ROG_IDLEX	ROG_STREX	ROG_HOTSOAK	ROG_RUNLOSS	ROG_DIURN	TOG_RUNEX	TOG_IDLEX	TOG_STREX	TOG_HOTSOAK	TOG_RUNLOSS	TOG_DIURN	NH3_RUNEX	CO_RUNEX	CO_IDLEX	CO_STREX	SOx_RUNEX	SOx_IDLEX	SOx_STREX
San Francisco	2027	HHDT	Aggregate	Aggregate	Diesel	1349.025231	101685.5171	101685.5171	0	12671.4416	3.205334022	37.42008201	3.222123784	0.024675006	0.039011788	0	0.008621638	0.033878232	0.025790701	0.040775728	0	0.034486551	0.096794948	1889.539977	6313.750428	0	0.001493368	0.124054022	0	0.297697793	0.994733951	0	0.032151798	2.670849083	0	0	0	0	0.036602375	3.040558373	0	0	0	0	0.194147785	0.127300163	36.49328617	0	0.017892814	0.059787441	0
San Francisco	2027	HHDT	Aggregate	Aggregate	Gasoline	1.448837512	73.34362942	73.34362942	0	28.98834093	7.288113052	0	0.147964683	0.002277718	0	0.001199621	0.005000001	0.033249697	0.002477226	0	0.001304697	0.020000006	0.094999135	2314.218144	0	58.65053124	0.194526319	0	7.50447E-05	0.217240398	0	0.003265365	1.119129401	0	0.000406703	0.110070048	0.963196413	7.120712232	1.633030995	0	0.000445289	0.110070048	0.963196413	7.120712232	0.045	40.37421626	0	6.785163279	0.022878402	0	0.00057982
San Francisco	2027	LDA	Aggregate	Aggregate	Diesel	418.4576539	10034.41594	10034.41594	0	1735.023208	0.145087508	0	0	0.015737468	0	0	0.002000001	0.003329609	0.016449047	0	0	0.008000002	0.009513169	250.7174316	0	0	0.001551565	0	0	0.039500633	0	0	0.032631054	0	0	0	0	0	0.037148286	0	0	0	0	0	0.0031	0.463436287	0	0	0.020375675	0	0
San Francisco	2027	LDT1	Aggregate	Aggregate	Diesel	2.12226489	20.32207681	20.32207681	0	5.891063406	1.194750365	0	0	0.22761149	0	0	0.002000001	0.004721019	0.237903071	0	0	0.008000002	0.013488626	437.4162287	0	0	0.013557946	0	0	0.068915105	0	0	0.291894603	0	0	0	0	0	0.332302599	0	0	0	0	0	0.0031	1.792868623	0	0	0.004144741	0	0
San Francisco	2027	LDT2	Aggregate	Aggregate	Diesel	324.3833365	10631.97802	10631.97802	0	1508.774676	0.043295577	0	0	0.005508497	0	0	0.002000001	0.004109684	0.005757567	0	0	0.008000002	0.011741955	330.6009267	0	0	0.001048889	0	0	0.052086311	0	0	0.022581956	0	0	0	0	0	0.025708055	0	0	0	0	0	0.0031	0.239307924	0	0	0.003132612	0	0
San Francisco	2027	LHDT1	Aggregate	Aggregate	Diesel	1834.326849	81069.52848	81069.52848	0	23073.52418	0.648568541	1.437574021	0	0.020636495	0.026645735	0	0.003000001	0.027300008	0.021569585	0.027850537	0	0.012000003	0.078000022	621.3526587	123.2447266	0	0.005174628	0.005098128	0	0.097894364	0.019417257	0	0.111406702	0.109759705	0	0	0	0	0.156829123	0.124954127	0	0	0	0	0.204149051	0.263761156	0.909745076	0	0.005887632	0.001167806	0
San Francisco	2027	LHDT2	Aggregate	Aggregate	Diesel	1076.831078	43893.79917	43893.79917	0	13545.18031	0.663670044	1.613866653	0	0.022777469	0.026839214	0	0.003000001	0.031850009	0.023807365	0.028052764	0	0.012000003	0.091000026	746.6330148	202.5616709	0	0.005496026	0.005098128	0	0.117632336	0.031913674	0	0.118326213	0.109759705	0	0	0	0	0.134706526	0.124954127	0	0	0	0	0.204910623	0.283037101	0.909745076	0	0.007074727	0.001919375	0
San Francisco	2027	LDA	Aggregate	Aggregate	Gasoline	123690.0398	4420178.822	4420178.822	0	573699.0431	0.03369265	0	0.231330199	0.001367796	0	0.001805849	0.002000001	0.003233179	0.001487603	0	0.001964026	0.008000002	0.009237654	273.3751038	0	65.64065397	0.002206573	0	0.061127035	0.004156052	0	0.030269777	0.008139262	0	0.279848119	0.084082162	0.225201615	1.324053362	0.011876793	0	0.306398465	0.084082162	0.225201615	1.324053362	0.037980439	0.677474147	0	2.87600279	0.007027591	0	0.000648925
San Francisco	2027	LDT1	Aggregate	Aggregate	Gasoline	12342.89567	367715.9729	367715.9729	0	55177.45872	0.071136595	0	0.3087366	0.001728588	0	0.002183359	0.002000001	0.004174927	0.001879997	0	0.002374602	0.008000002	0.011928362	331.4839315	0	80.28410153	0.004118055	0	0.079524262	0.006274486	0	0.034563607	0.017343931	0	0.391143647	0.127298715	0.351738614	1.988443873	0.025308224	0	0.428253059	0.127298715	0.351738614	1.988443873	0.036130969	1.01916852	0	3.880889841	0.003277056	0	0.00079369
San Francisco	2027	LDT2	Aggregate	Aggregate	Gasoline	67650.35726	2297095.068	2297095.068	0	316799.1703	0.046049074	0	0.273827839	0.001461889	0	0.001839314	0.002000001	0.004025835	0.001589938	0	0.002000422	0.008000002	0.011502386	338.0342319	0	80.70044591	0.002573267	0	0.068558853	0.004778739	0	0.033331019	0.009522813	0	0.310588784	0.069747537	0.188407191	1.139439308	0.013895667	0	0.340555624	0.069747537	0.188407191	1.139439308	0.038644525	0.748689076	0	3.117205168	0.003341813	0	0.000797806
San Francisco	2027	LHDT1	Aggregate	Aggregate	Gasoline	4945.315318	190933.66	190933.66	0	73677.82972	0.098009689	0.03401281	0.571461198	0.00130463	0	0.000201808	0.002000001	0.002730008	0.001418905	0	0.000219485	0.008000002	0.078000022	828.9418967	116.5962506	25.83741639	0.004933499	0.109490826	0.028770519	0.005737314	0.003014168	0.049315995	0.023572258	0.390058666	0.137925884	0.035677272	0.154233548	2.004753013	0.034396584	0.56917269	0.151011482	0.035677272	0.154233548	2.004753013	0.044991854	0.898738814	3.766370559	3.287060253	0.008194935	0.001152673	0.000255429
San Francisco	2027	LHDT2	Aggregate	Aggregate	Gasoline	609.7712474	22774.5641	22774.5641	0	9084.683027	0.099575395	0.033073052	0.568447308	0.001206833	0	0.000161348	0.002000001	0.031850009	0.001312541	0	0.000175481	0.008000002	0.091000026	937.1881842	134.8449869	25.05235906	0.003657942	0.10629232	0.027633782	0.00633875	0.002866637	0.047687713	0.015854797	0.379465371	0.131036092	0.034681187	0.185014775	1.987051102	0.023135283	0.553714978	0.143468027	0.034681187	0.185014775	1.987051102	0.044995477	0.727509729	3.770968892	3.209761912	0.009265059	0.001133308	0.000247668
																																												0.228919149	1.688747932					